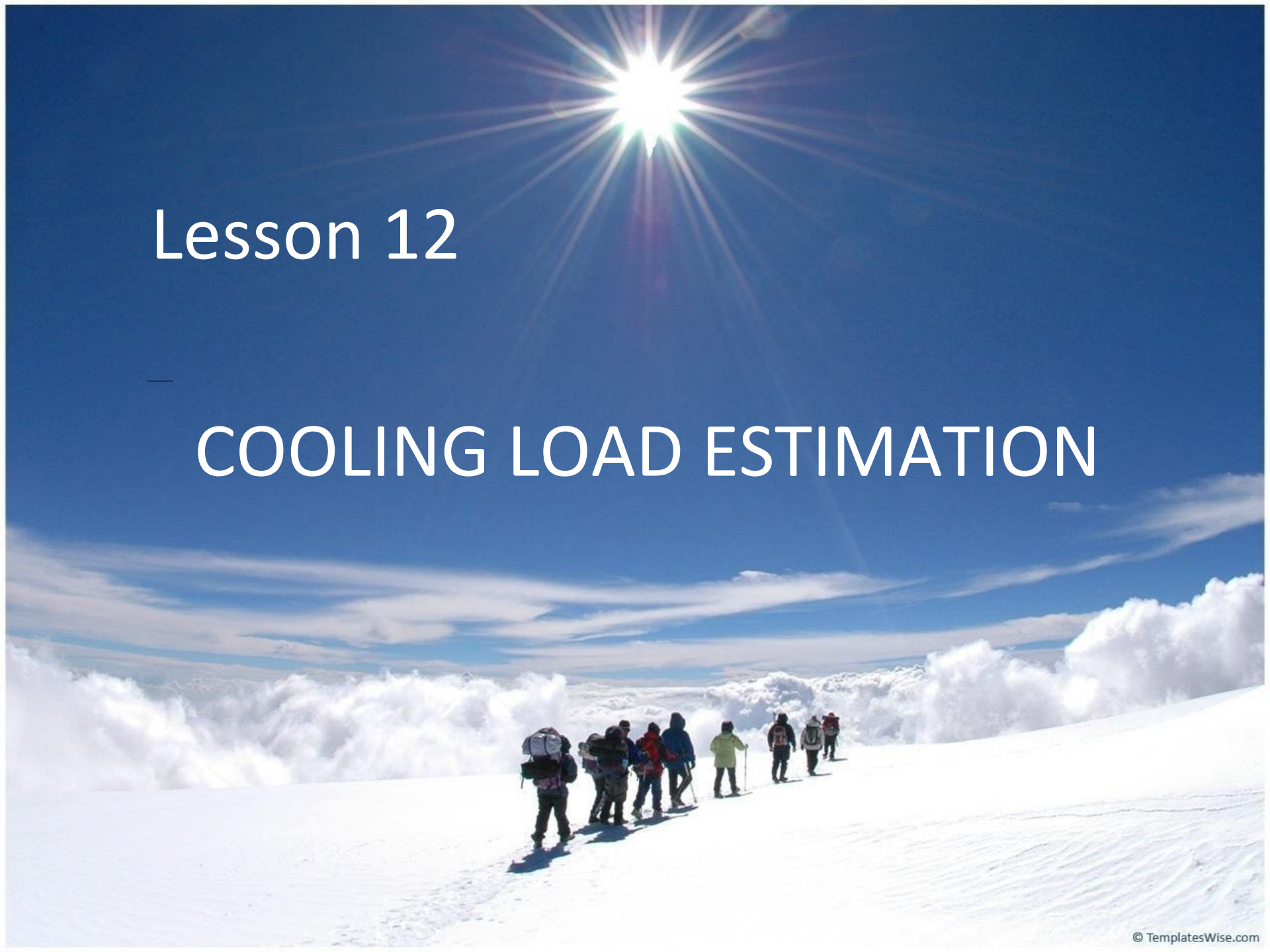

Lesson 12
COOLING LOAD ESTIMATION
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Introduction

- Accurate load estimate is needed
  - System selection
  - Equipment selections for fans, coils, and pumps,
  - Duct and electrical feeder sizing; and water piping design.
- Provide properly sized equipment – efficient operation

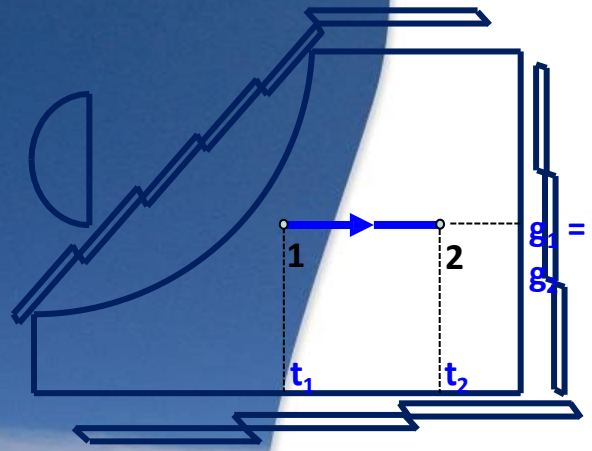
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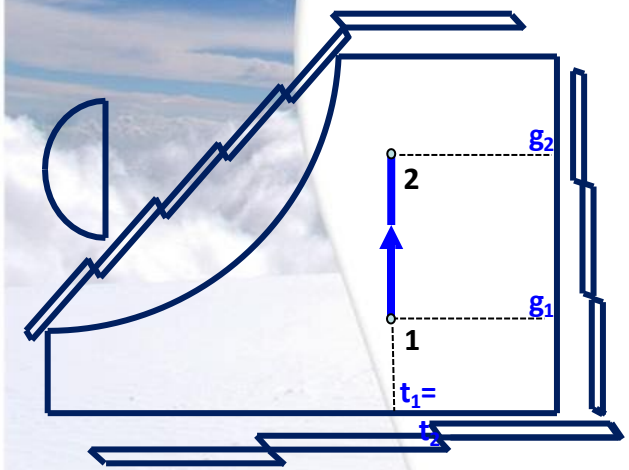
Definitions

- Space (or Room) Cooling Load - rate of heat removal from space required to maintain a constant space temp at a given time.
- Space (or Room) Heat Gain - instantaneous rate at which heat enters into and/or is generated within a space at a given time.

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Definitions





- Sensible heat gain : increase in enthalpy that causes a rise in room DB temp. and keeps moisture content unchanged.
- Latent heat gain : increase in enthalpy that causes a rise in moisture content (and RH) in a room and keeps room DB temp unchanged.

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Heat Gains

- Types of Heat Gains
  - Sensible
  - Latent
- Sources of Heat Gains
  - External
  - Internal

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Sources of Heat Gains

External

- Solar heat gain thru' windows (Sensible)
- Solar heat gain thru' opaque structures such as walls, roofs, etc (Sensible)
- Transmission heat gains thru windows and opaque structures due to air temperature difference (Sensible)
- Infiltration heat gain (Sensible & Latent)

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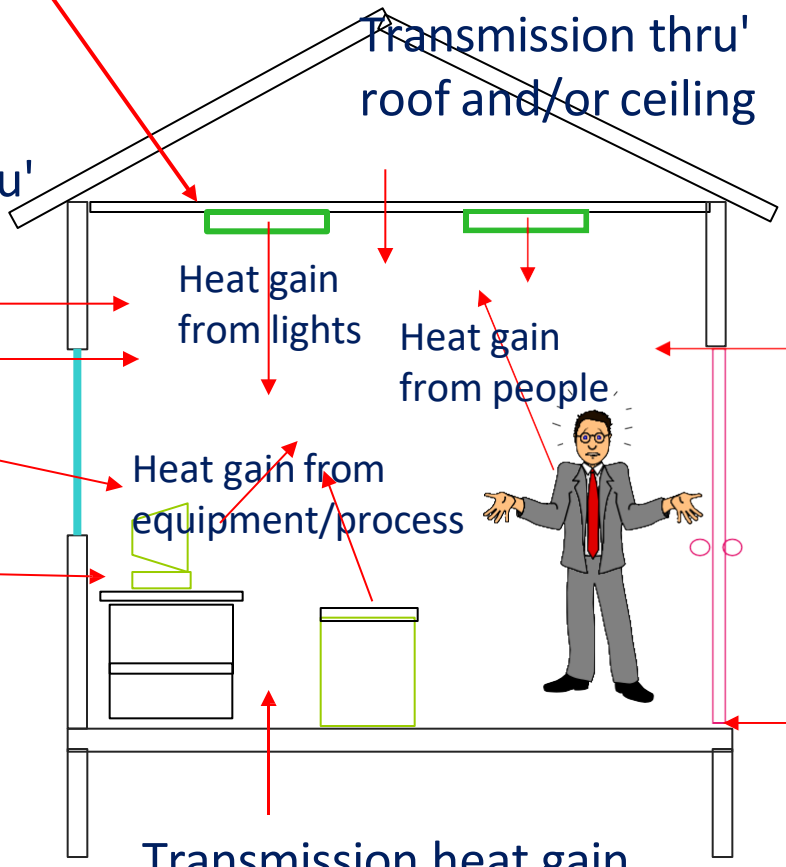
Sources of Heat Gains

Internal

- Lighting (Sensible)
- Occupancy (Sensible and Latent)
- Process or Equipment (Sensible and/or Latent)

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Sources of Heat Gains



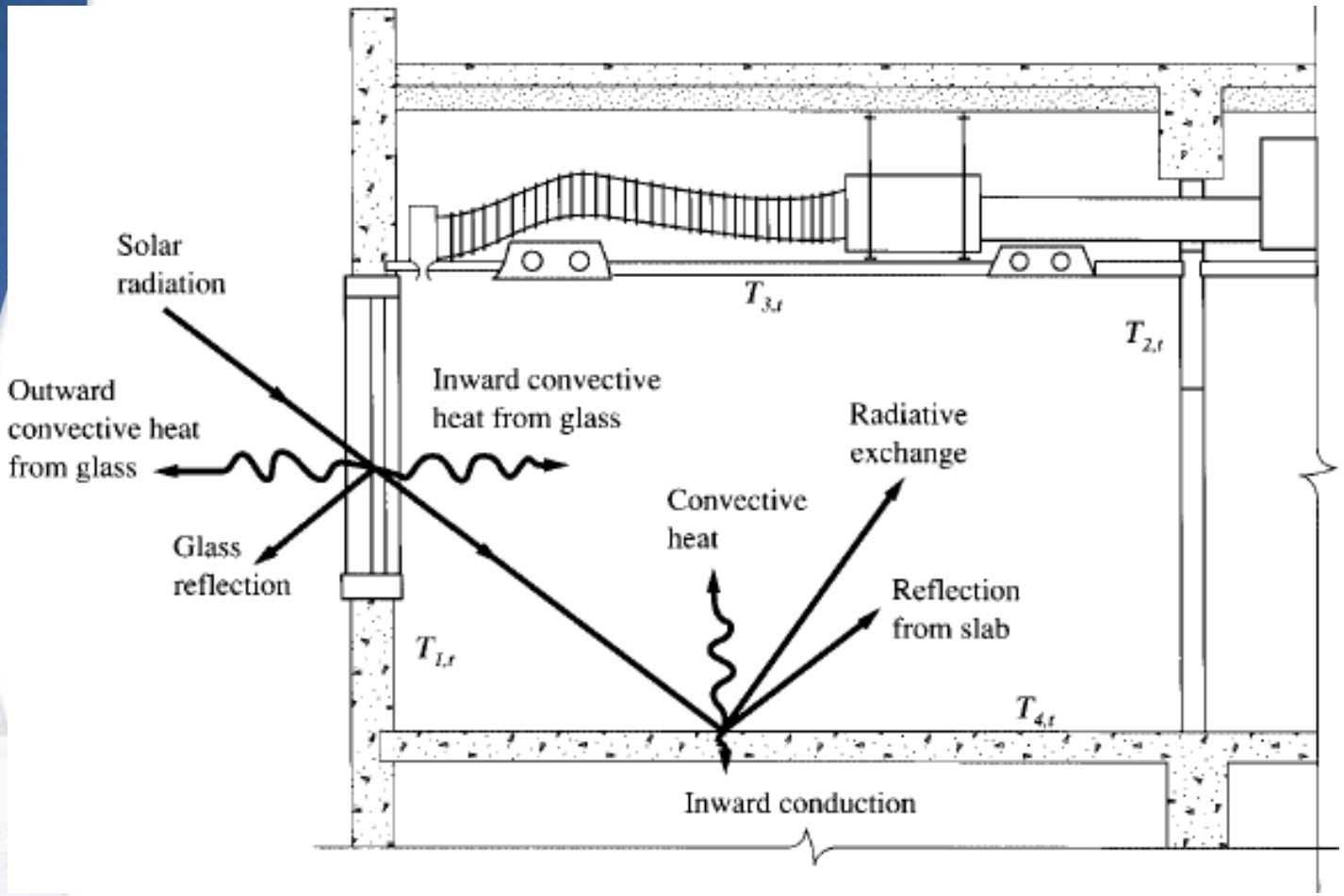
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Differences between Cooling Load and Heat Gain

- Space heat gains at any given time does not necessarily equal space cooling load for space at that same time. The reasons are:
  - *time lag between radiant heat gain and cooling load*
  - *heat storage by building materials and furniture*

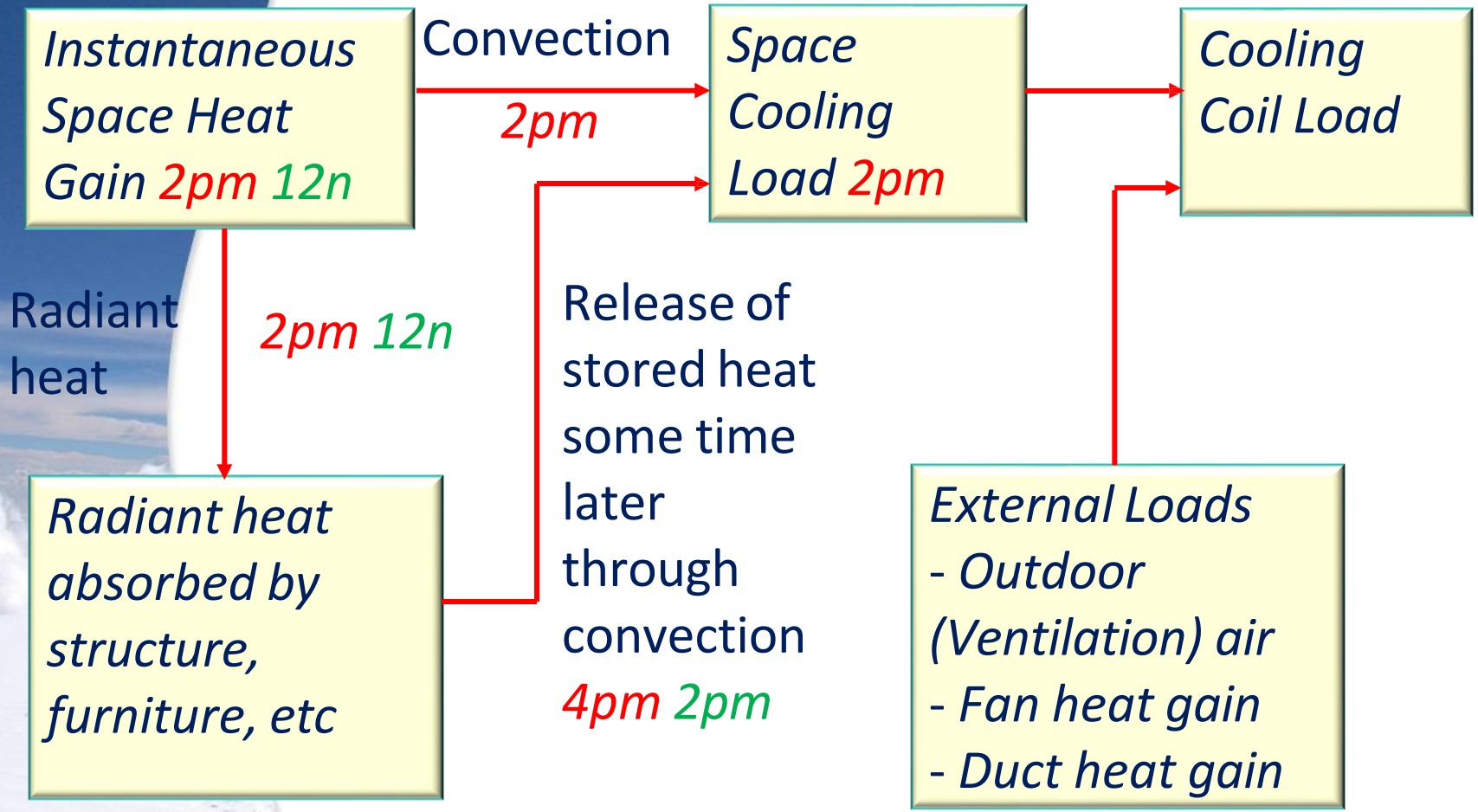
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Convective and Radiant Heat in a Conditioned Space



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Space Heat Gain & Cooling Load & Cooling Coil Load



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Cooling Coil Load

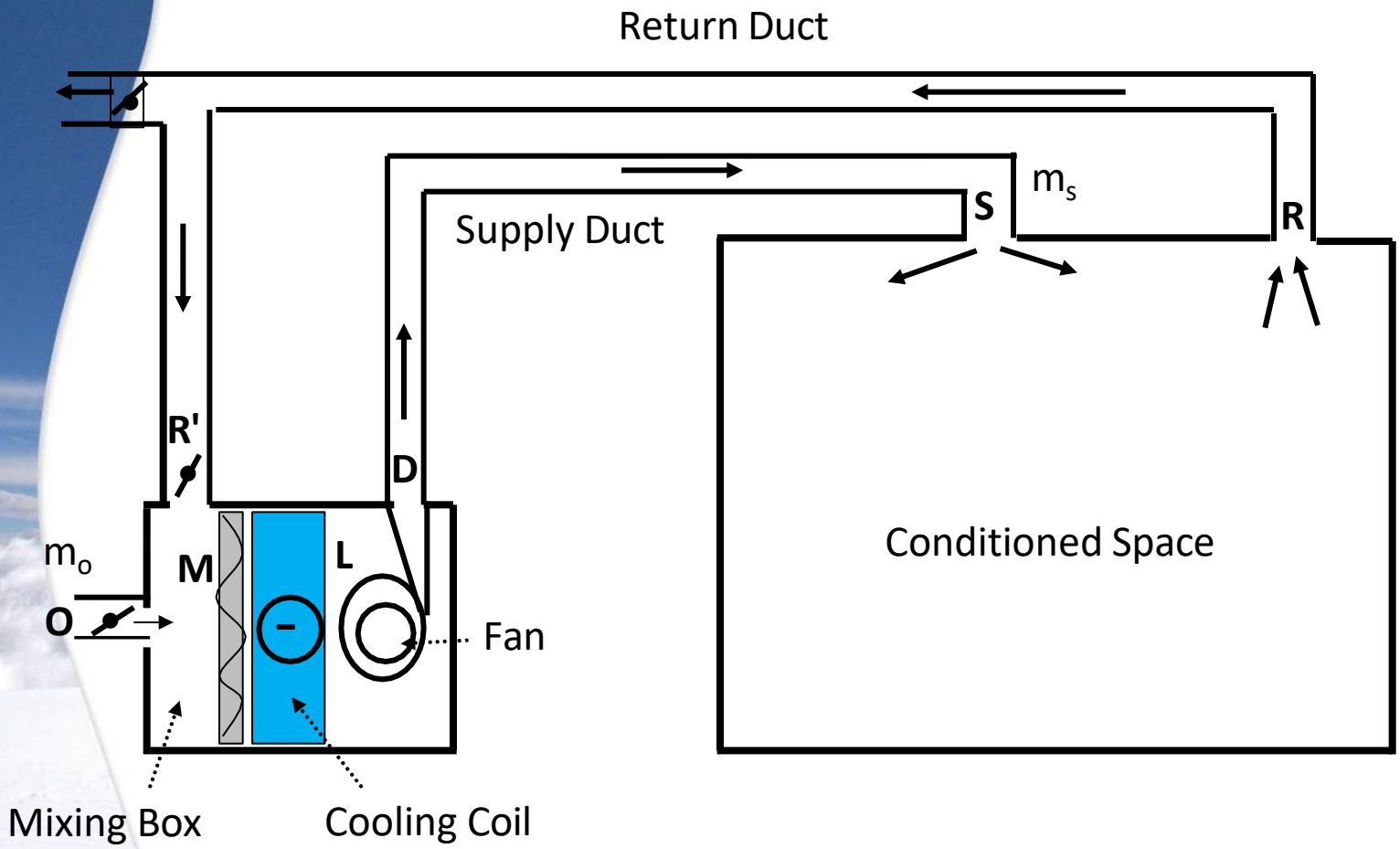
- Rate at which energy is removed at cooling coil that served one or more conditioned spaces.
- Sum of all the space cooling load plus any external loads.

External loads are heat gains due to:

- Supply and return air fans
- Supply air duct and return air duct
- Outdoor air introduced into the system

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Space Heat Gain & Cooling Load & Cooling Coil Load



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Cooling Load Estimation

Proper cooling load estimation requires gathering of sufficient data to obtain the level of accuracy required.

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Information for Cooling Load Estimation

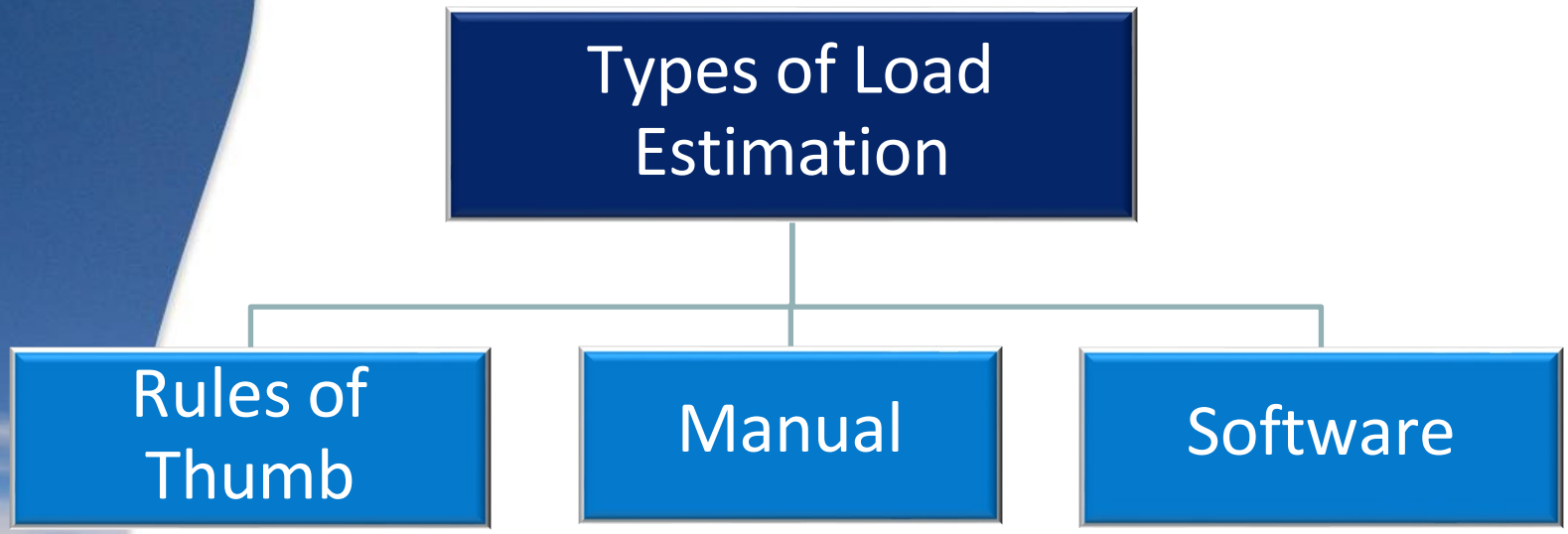
Location and elevation

Use of space

- Outdoor design condition
- Indoor design condition
- Occupancy of the entire building
- Orientation of the building
- Use of the spaces within the building
- Hours of operation
- Dimensions of walls, roofs, windows, doors
- Construction materials
- People occupancy, activity, diversity
- Lighting intensity and hours usage
- Motor and appliance sizes or kW, usage
- Ventilation needs

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Cooling Load Estimation



- W/sq m

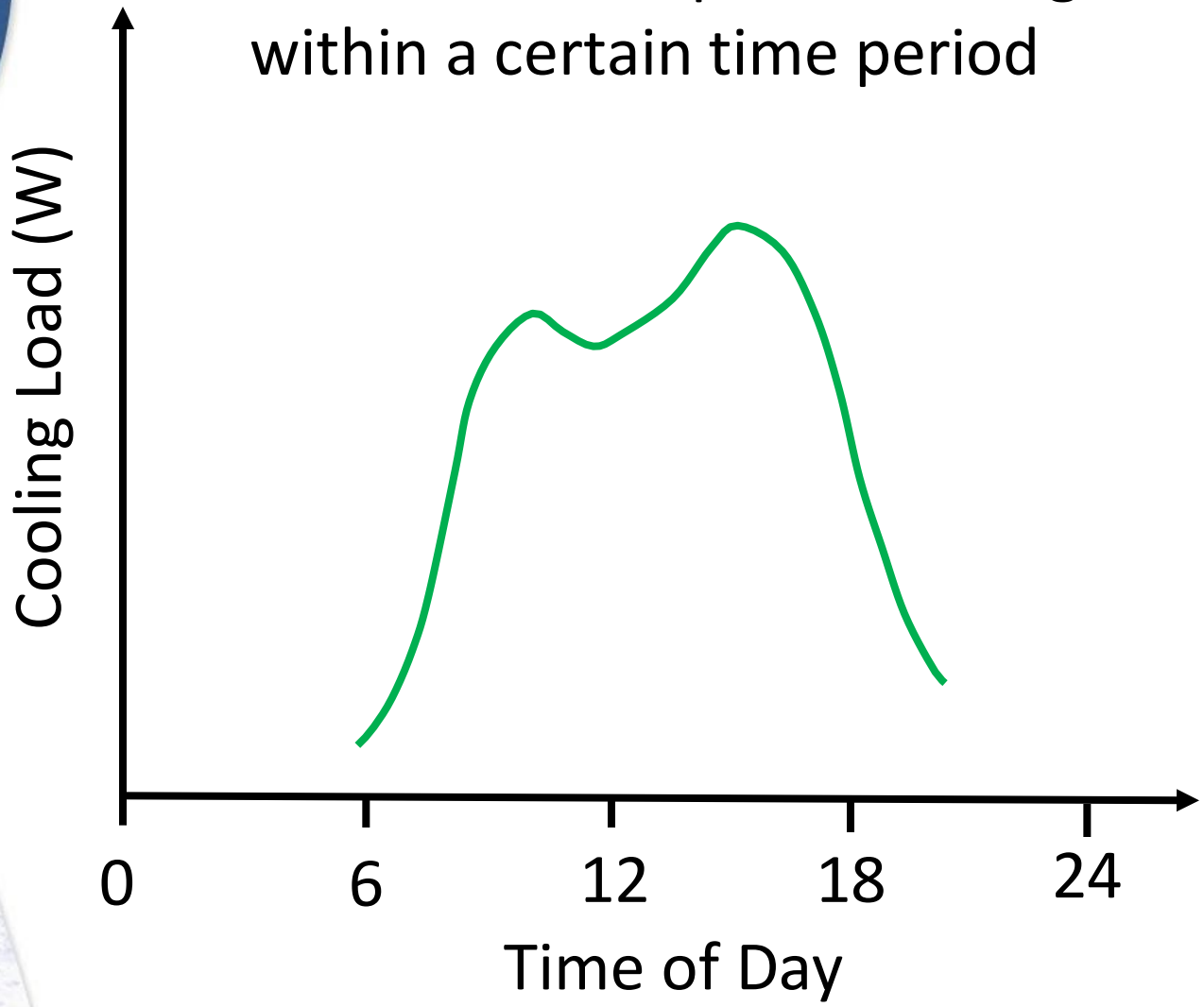
- Manual Calculations
- Spreadsheets

- Carrier E 20 II
- Trace 700

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Typical Floor Load Profile

Variation of space cooling load within a certain time period

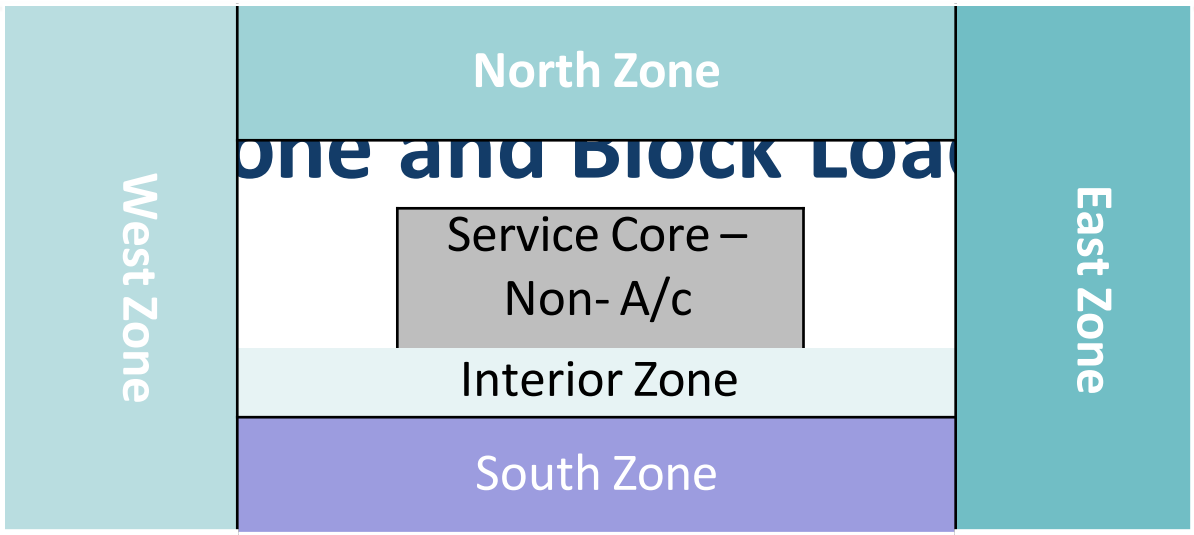


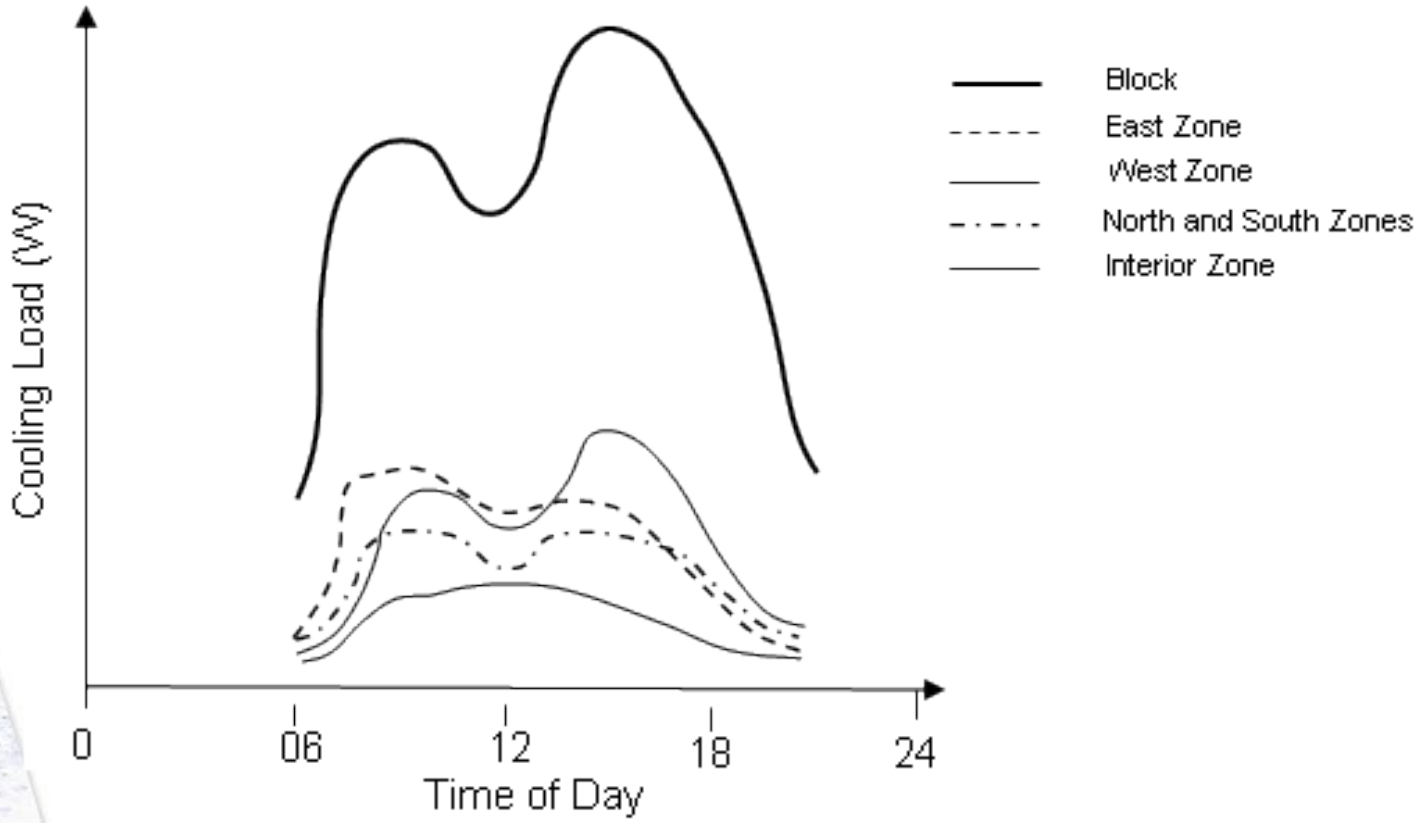
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Room, Zone and Block Loads

- Zone peak load
  - max space cooling load in load profile of control zone
  - max sum of zone cooling load components at a given time.
- Block load
  - max sum of zone cooling loads of various load profiles of control zones of a floor or building at the same time.
  - max sum of cooling load components in that space, room, floor, or building at a given time.

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Uses of Cooling Load Estimates

Load types are used for equipment sizing :

- cooling load and supply air flowrates for supply air outlets selection
- duct sizing
- fan coil unit and air handling unit supply air fan selection
- fan coil unit and air handling unit capacity selection
- chilled water pipe sizing
- condenser water pipe sizing
- refrigeration equipment selection

Solar (Sun) Time

- Time in hours where noon is time when sun is highest in sky.



oon phenomenon as seen at Kallang Wave Mall on March 23. It takes place twice a year in countries near the solar altitude is as close to or at 90 degrees. ST PHOTO: SHINTARO TAY

1.11pm, March 23: Look, no shadows

Elaine Lee

The Lahaina Noon phenomenon is a natural event in which objects standing upright, such as a stick or pole, will cast no shadow.

A Lahaina Noon phenomenon, where one's shadow appears minimal, took place at 1.11pm on March 23 in Singapore.

Also known as zero shadow day, it is a natural event in which objects standing upright, such as a stick or pole, will cast no shadow.

Professor Matthias Roth, who teaches geography at the National University of Singapore, said the phenomenon will also happen on Sept 19 at 12.58pm, when the solar altitude is as close to or at 90 degrees.

He noted that the phenomenon on March 23 comes about four days after the start of the spring astronomical season, which is determined by the position of the Earth in relation to the Sun.

As Singapore is not exactly on the Equator, the midday sun will be overhead only about four days after the astronomical spring equinox, Prof Roth added.

The astronomical spring equinox results in an equal amount of daylight and darkness.

"Due to historical reasons, Singapore's time is one hour ahead compared with the solar time. This means that solar noon does not occur until about 1pm, rather than noon, which also explains why sunrise and sunset times are observed at about 7am and 7pm respectively."

Science Centre Singapore's senior science educator Meng Hwee Lim said the Lahaina Noon phenomenon takes place twice a year in areas situated between the tropics of Cancer and Capricorn. Other places include Malaysia, Brazil, Hawaii, Thailand and Hong Kong.

For places near the Equator, it typically occurs near the astronomical spring equinox on March 21 and during the astronomical autumn equinox on Sept 21, added Mr Meng, who is from the centre's schools and professional development team.

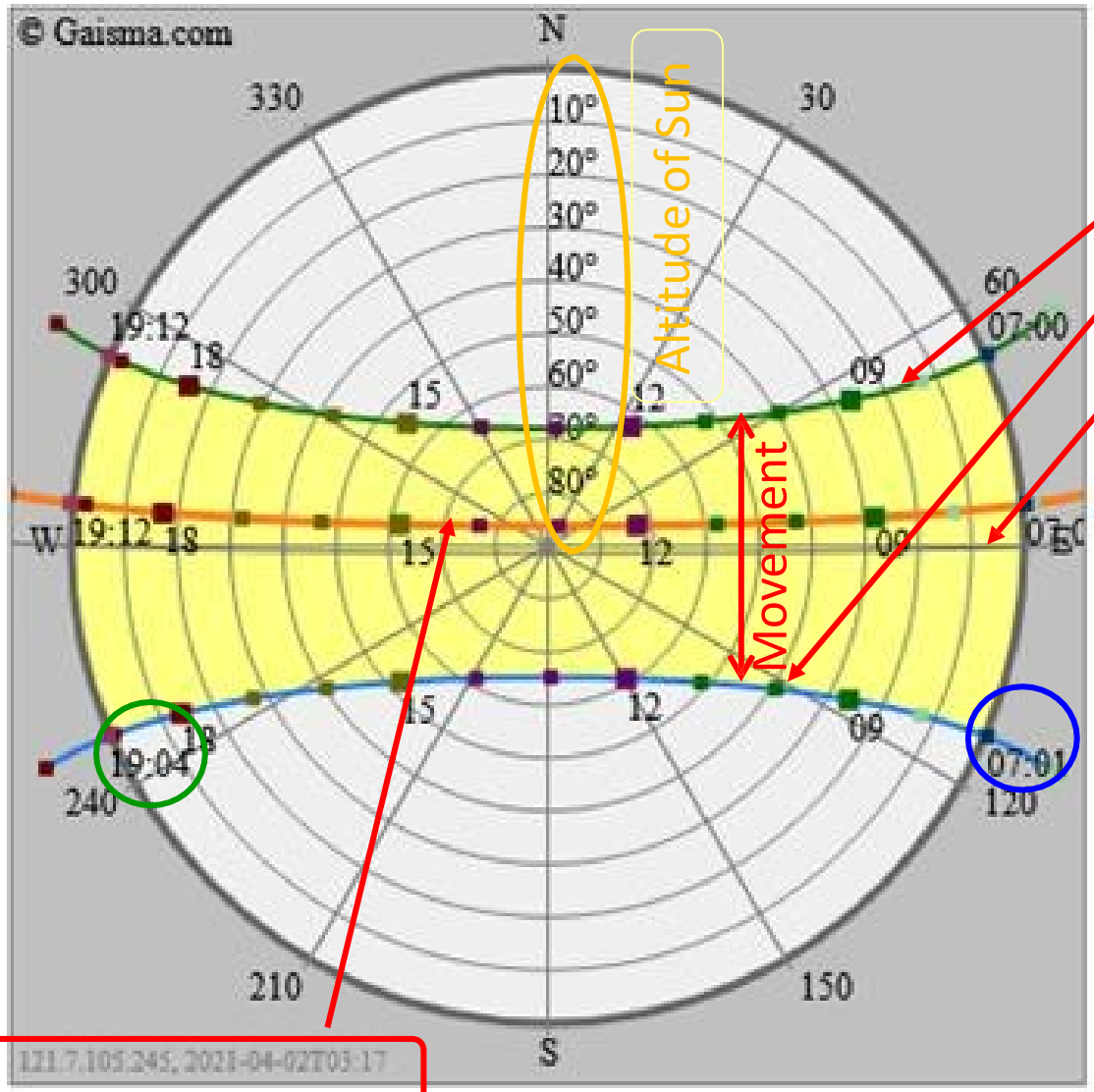
"The effect is expected to last for just a few minutes."

According to Malaysia's Ministry of Science, Technology and Innovation, the event will take place after the astronomical spring equinox in Putrajaya on March 27, and in Kuala Lumpur on March 28.

huienl@sph.com.sg

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Sun Path



Sun path
- Today
- June solstice
- December solstice
- Annual variation
- Equinox (March and September)

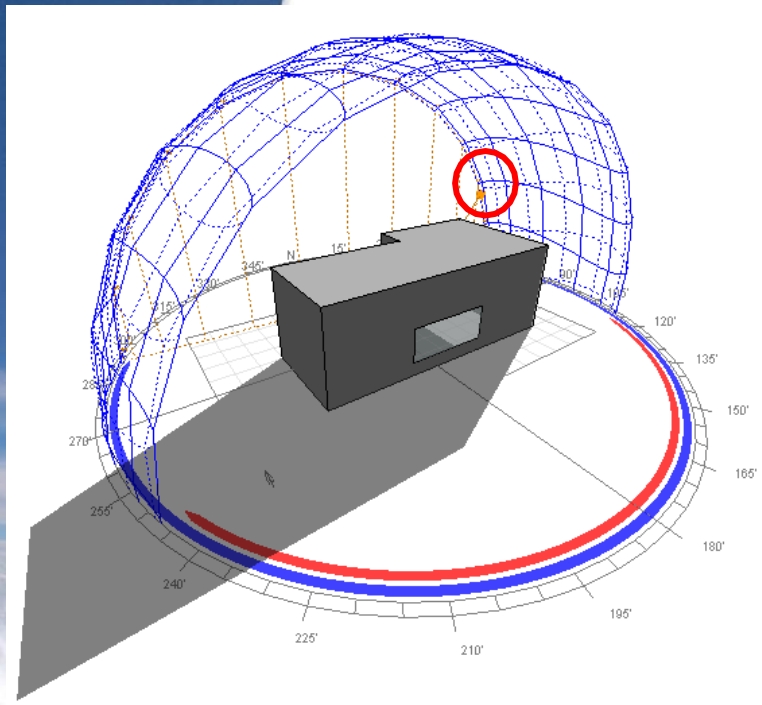
Sunrise/sunset
- Sunrise
- Sunset

Time
- 00-02
- 03-05
- 06-08
- 09-11
- 12-14
- 15-17
- 18-20
- 21-23

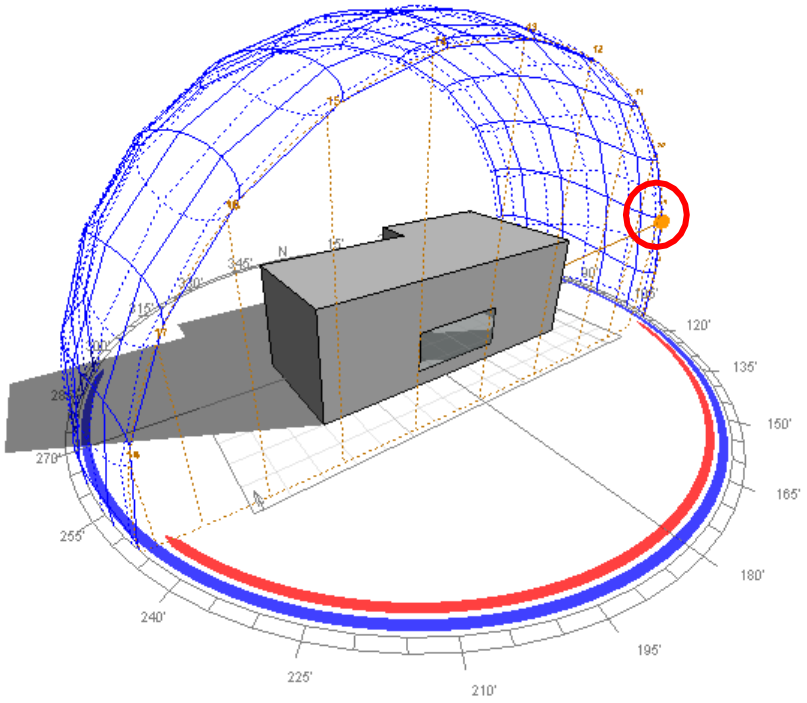
2 April 2021

http://www.gaisma.com/en/location/singapore.html

Sun Path Diagram



Singapore (15 June 8:45 AM)

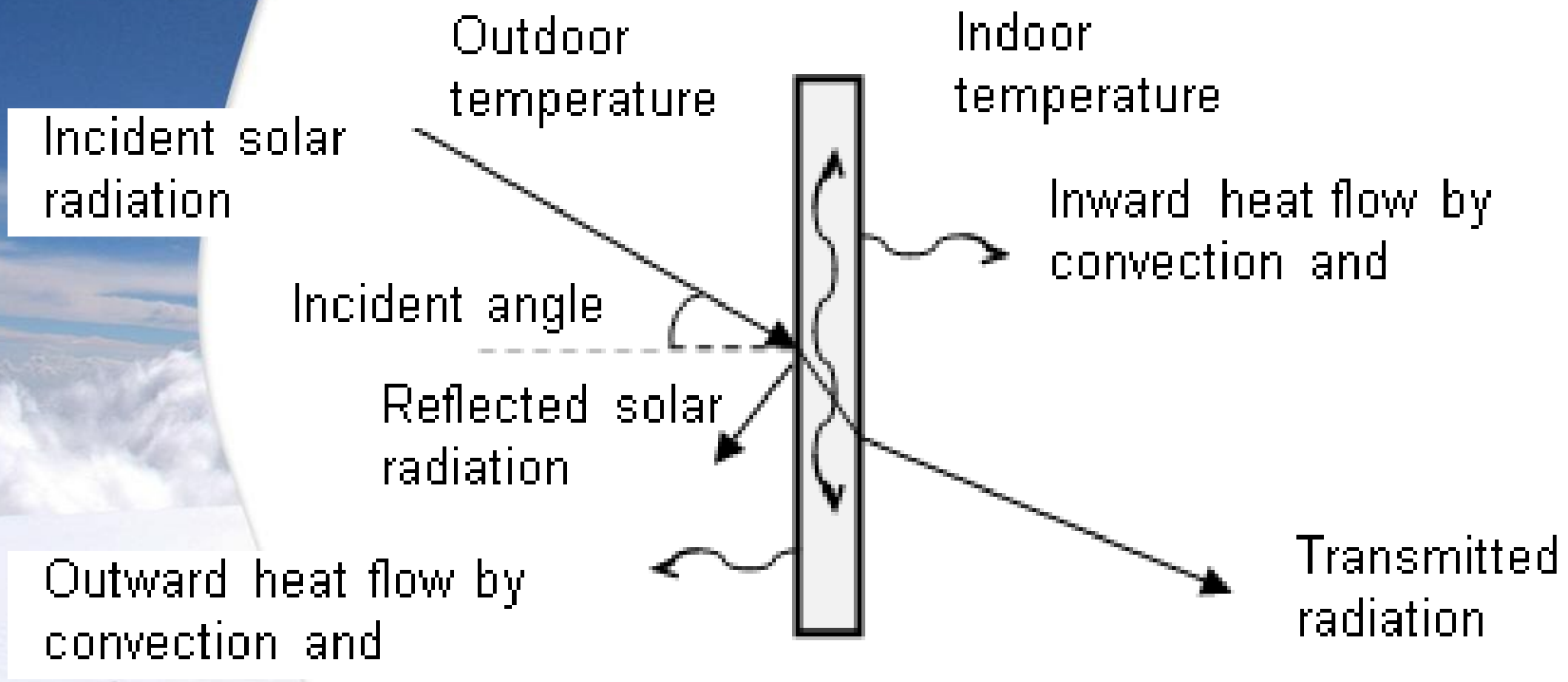


Singapore (15 Dec 8:45 AM)

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Cooling Load due to Glazing

- Cooling load due to solar radiation
- Cooling load due to heat conduction



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Cooling Load due to Solar Radiation through Vertical Glazing

$Qc_1 = A_g \times SHGF \times SC \times CLF$ Watts

Where: A_g = area of glazing, m^2

SHGF = Solar heat gain factor, W/m^2 (Table 4.1)

SC = Shading coefficient of glass (Table 4.2)

CLF = Cooling load factor (Table 4.3)

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Table 4.1 – SHGF (Singapore)

Month	N	NNE/ NNW	NE/ NW	ENE/ WNW	E/W	ESE/ WSW	SE/ SW	SSE/ SSW	S	Hor
	Maximum Solar Heat Gain Factor (W/m²)									
Jan	107	107	268	547	727	793	745	586	399	922
Feb	111	113	403	636	766	776	667	458	234	958
Mar	120	260	524	693	755	701	540	282	123	953
Apr	204	417	607	702	690	577	370	123	119	893
May	332	506	648	695	637	485	254	117	117	837
Jun	385	532	654	682	607	442	209	118	118	808
Jul	341	503	637	680	620	469	247	120	120	824
Aug	217	416	594	681	667	554	353	125	123	870
Sep	124	252	500	658	718	669	519	278	127	919
Oct	113	117	392	613	734	743	644	444	229	936
Nov	108	108	269	536	710	780	735	578	392	911
Dec	104	104	220	496	693	792	769	627	459	891

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Table 4.2 – Shading Coefficient

Type of Glazing	SC
Uncoated Single Glazing	
6.4mm clear	0.89
6.4mm bronze	0.55
6.4mm green	0.74
6.4mm gray	0.43
Reflective Single Glazing	
6.4mm SS on CLR 8%	0.22
6.4mm SS on CLR 14%	0.29
6.4mm SS on GRN 14%	0.29
Uncoated Double Glazing	
6.4mm CLR CLR	0.81
6.4mm BRZ CLR	0.59
6.4mm GRN CLR	0.54
6.4mm GRY CLR	0.51

Type of Glazing	SC
Reflective Double Coating	
6.4mm CLR 8% CLR	0.15
6.4mm CLR 14% CLR	0.2
6.4mm GRN 14% CLR	0.18
Low-e Double Glazing e=0.2 on surface 2	
6.4mm LE CLR	0.7
Low-e Double Glazing e=0.2 on surface 3	
6.4mm CLR LE	0.75
6.4mm BRZ LE	0.52
6.4mm GRN LE	0.48
6.4mm GRY LE	0.37
Low-e Double Glazing e=0.1 on surface 1	
6.4mm LE CLR	0.59

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Table 4.3 - CLF

CLF for Glass without Internal Shading, in North Latitude having Carpeted Floors												
		Solar Time										
Exposure	**Bldg Weight**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
N	L	0.73	0.74	0.81	0.88	0.95	0.98	0.98	0.94	0.88	0.79	0.79
	M	0.69	0.69	0.77	0.84	0.91	0.94	0.95	0.91	0.86	0.79	0.79
	H	0.64	0.64	0.71	0.77	0.83	0.87	0.88	0.85	0.81	0.75	0.76
NE	L	0.83	0.88	0.67	0.44	0.32	0.28	0.26	0.24	0.22	0.19	0.15
	M	0.78	0.82	0.70	0.58	0.40	0.29	0.26	0.24	0.21	0.19	0.16
	H	0.71	0.72	0.59	0.40	0.30	0.27	0.26	0.25	0.23	0.20	0.17
E	L	0.76	0.91	0.90	0.75	0.51	0.30	0.22	0.18	0.16	0.13	0.11
	M	0.72	0.86	0.84	0.71	0.48	0.30	0.24	0.21	0.18	0.16	0.13
	H	0.66	0.76	0.74	0.63	0.43	0.29	0.24	0.22	0.20	0.18	0.15
SE	L	0.58	0.81	0.93	0.93	0.81	0.59	0.37	0.27	0.21	0.18	0.14
	M	0.55	0.77	0.88	0.87	0.76	0.56	0.37	0.29	0.24	0.20	0.16
	H	0.51	0.69	0.78	0.78	0.68	0.51	0.35	0.29	0.25	0.22	0.19
S	L	0.15	0.23	0.39	0.62	0.82	0.94	0.93	0.80	0.59	0.38	0.26
	M	0.14	0.22	0.38	0.59	0.78	0.88	0.88	0.76	0.57	0.38	0.28
	H	0.15	0.21	0.35	0.54	0.70	0.79	0.79	0.69	0.52	0.37	0.29

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Table 4.3 - CLF

CLF for Glass without Internal Shading, in North Latitude having Carpeted Floors												
		Solar Time										
Exposure	**Bldg Weight**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
SW	L	0.09	0.13	0.16	0.19	0.23	0.39	0.62	0.82	0.94	0.94	0.81
	M	0.09	0.13	0.16	0.19	0.22	0.38	0.60	0.78	0.89	0.89	0.77
	H	0.11	014	0.16	0.18	0.21	0.35	0.55	0.71	0.80	0.79	0.69
W	L	0.07	0.10	0.13	0.15	0.16	0.18	0.31	0.55	0.78	0.92	0.93
	M	0.07	0.10	0.13	0.14	0.16	0.17	0.30	0.53	0.74	0.87	0.88
	H	0.09	0.11	0.13	0.15	0.16	0.17	0.28	0.49	0.67	0.78	0.79
NW	L	0.09	0.14	0.17	0.20	0.22	0.23	0.24	0.31	0.53	0.78	0.92
	M	0.10	0.13	0.17	0.19	0.21	0.22	0.23	0.30	0.52	0.75	0.88
	H	0.11	0.14	0.17	0.19	0.20	0.21	0.22	0.28	0.48	0.68	0.79
Horizontal	L	0.25	0.45	0.64	0.80	0.91	0.97	0.97	0.91	0.80	0.64	0.44
	M	0.24	0.43	0.60	0.75	0.86	0.92	0.92	0.87	0.77	0.63	0.45
	H	0.25	0.41	0.56	0.68	0.77	0.83	0.80	0.71	0.59	0.44	0.28

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Space

Cooling Load due to Transmission through Glazing

- $Q_{c2} = A_g \times U_g \times CLTD$

Where: U_g = thermal transmittance of glass, W/m^2K

CLTD = cooling load temperature difference, K or °C

Since glass is thin heat storage of glass can be neglected.

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Table 4.4 - CLTD

Solar Time, h	CLTD (°C)	Solar Time, h	CLTD (°C)
1	1	13	7
2	0	14	7
3	-1	15	8
4	-1	16	8
5	-1	17	7
6	-1	18	7
7	-1	19	6
8	0	20	4
9	1	21	3
10	2	22	2
11	4	23	2
12	5	24	1

Values based on inside temperature of 25.5°C with an outdoor maximum of 35°C with outdoor daily range of 11.6°C. Approximately correct for other outdoor maximum 33 to 39°C and other outdoor daily ranges 9 to 19°C provided outdoor daily average remains approximately 29.5°C. Correction for other conditions corrected CLTD = CLTD + (25.5 – t_r) + (t_m – 29.4) where t_r = inside temperature and t_m = maximum outdoor temperature – (daily range/2).

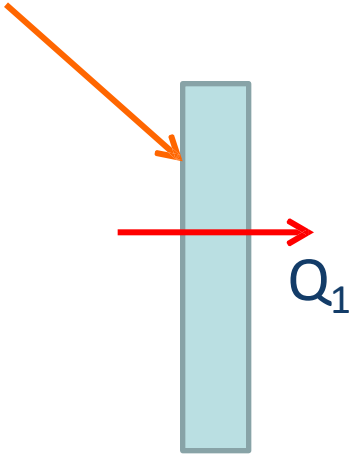
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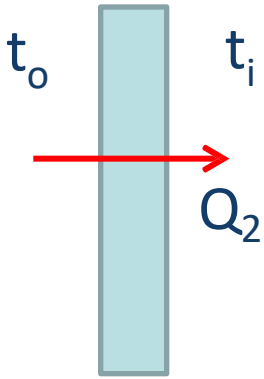
Cooling Load due to External Walls and Roofs

- Consists of two parts:

SOLAR heat gain due to solar radiation incident on external surface of structure.

TRANSMISSION heat gain due to indoor-to-outdoor air temperature difference

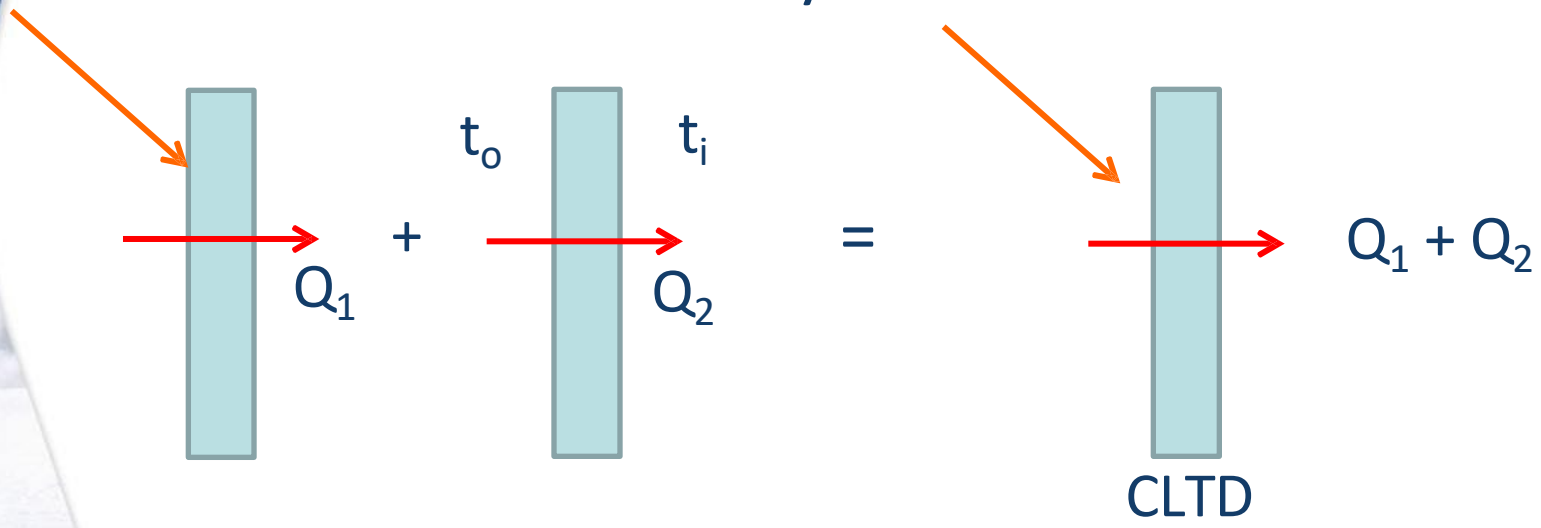




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Cooling Load through External Wall or Roof

- CLTD is used to estimate cooling load through external walls and roofs. Effects of time lag and heat storage incorporated in CLTD data already.



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Cooling Load through External Wall or Roof

$Q_{c3} = A_w \times U_w \times CLTD$ Watts

Where: U_w = thermal transmittance for roof or wall (W/m^2K)

CLTD = cooling temperature difference, K or °C

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CLTD for Sunlit Group D Walls – Mar/Sep, 0°N

Cooling Load Temperature Differences (CLTD) for Sunlit Group D Walls											
	Solar Time										
Expo-sure	7	8	9	10	11	12	13	14	15	16	17
N	1.7	1.7	1.7	1.7	1.7	2.2	2.8	3.9	5.0	5.6	6.7
NE	4.4	5.0	6.1	8.3	4.4	11.7	12.8	13.3	13.3	13.9	13.9
E	3.9	4.4	6.1	8.9	11.7	14.4	16.1	17.2	17.8	17.8	17.2
SE	2.8	2.8	3.9	5.6	7.8	10.6	12.8	14.4	15.6	16.1	16.1
S	0.0	-0.6	-1.1	-1.1	-0.6	0.6	1.7	4.4	6.7	8.9	10.6
SW	5.0	3.9	3.3	2.8	2.8	2.8	3.9	5.0	7.2	10.0	13.3
W	5.6	4.4	3.9	3.3	3.3	3.3	3.9	4.4	6.1	8.3	11.7
NW	6.1	5.6	3.9	4.4	4.4	5.0	5.6	6.1	7.2	8.3	10.6

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CLTD for Sunlit Group D Walls – June, 0°N

Cooling Load Temperature Differences (CLTD) for Sunlit Group D Walls											
	Solar Time										
Expo-sure	7	8	9	10	11	12	13	14	15	16	17
N	10.0	10.0	10.0	10.0	10.0	10.6	11.1	12.2	13.3	13.9	15.0
NE	6.7	7.2	8.3	10.6	6.7	13.9	15.0	15.6	15.6	16.1	16.1
E	2.8	3.3	5.0	7.8	10.6	13.3	15.0	16.1	16.7	16.7	16.1
SE	-0.6	-0.6	0.6	2.2	4.4	7.2	9.4	11.1	12.2	12.8	12.8
S	0.0	-0.6	-1.1	-1.1	-0.6	0.6	2.2	4.4	6.7	8.9	10.6
SW	1.7	0.6	0.0	-0.6	-0.6	-0.6	0.6	1.7	3.9	6.7	10.0
W	5.6	4.4	3.9	3.3	3.3	3.3	3.9	4.4	6.1	8.3	11.7
NW	8.3	7.8	7.2	6.7	6.7	7.2	7.8	8.3	9.4	10.6	12.8

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U-values for Walls

Table -1
U-values for Walls

Wall description	Group Number	Description of construction	Weight, lb/ft²	U value, Btu/h·ft²·°F	Code numbers for layers
4-in face brick + brick	C	Airspace + 4-in face brick	83	0.358	A0 A2 B1 A2 E0
	D	4-in common brick	90	0.415	A0 A2 C4 E1 E0
	C	1-in insulation or airspace + 4-in common brick	90	0.174-0.301	A0 A2 C4 B1/B2 E1 E0
	B	2-in insulation + 4-in common brick	88	0.111	A0 A2 B3 C4 E1 E0
	B	8-in common brick	130	0.302	A0 A2 C9 E1 E0
	A	Insulation or airspace + 8-in common brick	130	0.154-0.243	A0 A2 C9 B1/B2 E1 E0
4-in face brick + (lightweight or heavyweight concrete block)	E	4-in block	62	0.319	A0 A2 C2 E1 E0
	D	Airspace or insulation + 4-in block	62	0.153-0.246	A0 A2 C2 B1/B2 E1 E0
	D	8-in block	70	0.274	A0 A2 C7 A6 E0
	C	Airspace or 1-in insulation + 6-in or 8-in block	73-89	0.221-0.275	A0 A2 B1 C7/C8 E1 E0
	B	2-in Insulation + 8-in block	89	0.096-0.107	A0 A2 B3 C7/C8 E1 E0
Heavyweight concrete wall + (finish)	E	4-in concrete	63	0.585	A0 A1 C5 E1 E0
	D	4-in concrete + 1-in or 2-in insulation	63	0.119-0.200	A0 A1 C5 B2/B3 E1 E0
	C	2-in insulation + 4-in concrete	63	0.119	A0 A1 B6 C5 E1 E0
	C	8-in concrete	109	0.49	A0 A1 C10 E1 E0
	B	8-in concrete + 1-in or 2-in insulation	110	0.115-0.187	A0 A1 C10 B5/B6 E1 E0
	A	2-in insulation + 8-in concrete	110	0.115	A0 A1 B3 C10 E1 E0
	B	12-in concrete	156	0.421	A0 A1 C11 E1 E0

Official Open

Space

Cooling Load due to Internal Walls and Partitions

Only the TRANSMISSION heat gain due to the difference in air temperatures.

$$Q_{c4} = A_w.U_w.(t_{nr} - t_i)$$

Where: A_w = area of partition/wall, m^2

U_w = thermal transmittance of partition/wall, W/m^2K

t_{nr} = air temperature in the next room, °C

Official Open

Cooling Load due to Internal Heat Sources

- Cooling Load due to People
- Cooling Load due to Lighting
- Cooling Load due to Equipment

Official Open

Cooling Load due to People

- Density of occupation - Typical values
  Offices - 6 - 10 m^2/person
  Restaurants – 1 - 2 m^2/person

- Sensible cooling load due to occupancy

  Q_{c5} = No. of occ. x sensible gain/person x CLF x DF

- Latent cooling load due to occupancy

  Q_{c6}' = No. of occ. x latent gain/person x CLF x DF

CLF = Cooling load factor by hour of occupancy

CLF = 1.0 with high density or 24-h occupancy and/or cooling off at night or during weekends

Official Open

Cooling Load due to People

Application	Summer Inside Design Conditions		Occupancy Density	Miscellaneous Load
Office (general)	24	50-55	9 (Perimeter Zone) 6 (Interior Zone)	25
Office (High tech)	24	50-55	9 (Perimeter Zone) 6 (Interior Zone)	50-70
Conference Room	24	55	2	25
Chinese Restaurant	22	60	1	10 sensible/person 10 latent/person (for meal)
Western Restaurant	22	60	1.5	10 sensible/person 10 latent/person (for meal)
Coffee Shop/Bar/ Lounge (not for meals)	24	60	1.5	10
Canteen/Food Plazas	22	60	1	10 sensible/person 10 latent/person (for meal)
Banquet Room	22	60	1	10 sensible/person 10 latent/person (for meal)
Kitchen	Spot cooling	-	-	-

Official Open

Cooling Load due to People

Application	Summer Inside Design Conditions		Occupancy Density	Miscellaneous Load
Retail Shop	24	55	2.5	40
Supermarket	24	55-60	12.5	5-10
Shopping arcade	24	55	2.5	25-40
Toilet/Changing Room (with A/C)	26	-	-	15-40
Cinema	24	55	1 or Number of Seats	-
Storeroom (With A/C	24-26	55	20	15-40
Computer Room (PCs/Terminals Mainly)	20	50	6-10	15-40
Computer Room (Mainframe mainly)	20	50	10-15	15-40
Hotel Guestroom	24	55	2 person/room or design accommodation capacity	500 W per room
Corridor	24	55-60	-	-
Main Entrance/Large Lobbies	24	55-60	-	-
Lift Lobby	24	55-60	-	-

Official Open

Cooling Load due to People

Application	Summer Inside Design Conditions		Occupancy Density	Miscellaneous Load
Health Club	24	60	8-10	-
Multi-function Room	24	55	1.5	40
Back-of-house Area	24-26	55-60	4	20
Disco/Dancing Hall/Ballroom	22	55	1	40
Karaoke Room	24	55	1.5	40
Banking Hall	24	55	1.5	10
Bank Vault	24	55	-	-
Hospital Ward Room	24	55	8-10	20
Hospital Operating Theatre	20-24	50-60	4-6	40-60
Hospital Consultation Room	24	55	4.5	10
Clinic	24	55	1.5	10
Pantry Room	26	60	-	40

Official Open

Applications		Sensible (s) and latent (l) heat emissions, W, at the stated dry bulb temperature, °C									
Degree of Activity	Typical	Total	15		20		22		24		26	
			(s)	(l)	(s)	(l)	(s)	(l)	(s)	(l)	(s)	(l)
Seated at rest	Theatre, hotel lounge	115	100	15	90	25	80	35	75	40	65	50
Light work	Office, restaurant*	140	110	30	100	40	90	50	80	60	70	70
Walking slowly	Department Store, bank	160	120	40	110	50	100	60	85	75	75	85
Light bench work	Factory	235	150	85	130	105	115	120	100	135	80	155
Medium work	Factory, dance hall	265	160	105	140	125	125	140	105	160	90	175
Heavy work	Factory	440	220	220	190	250	165	275	135	305	105	335

* **For restaurants serving hot meals, add 10 W sensible/person and 10 W latent/person for food.**

Official Open

Table 1 Representative Rates at Which Heat and Moisture Are Given Off by Human Beings in Different States of Activity

Degree of Activity	Location	Total Heat, W: Adult Male	Total Heat, W: Adjusted, M/F[a]	Sensible Heat, W	Latent Heat, W	% Sensible Heat that is Radiant[b]: Low *V*	% Sensible Heat that is Radiant[b]: High *V*
Seated at theater	Theater	114	103	72	31	60	27
Seated, very light work	Offices, hotels, apartments	132	117	72	45		
Moderately active office work	Offices, hotels, apartments	139	132	73	59		
Standing, light work; walking	Department store; retail store	161	132	73	59	58	38
Walking, standing	Drug store, bank	161	147	73	73		
Sedentary work	Restaurant[c]	144	161	81	81		
Light bench work	Factory	235	220	81	139		
Moderate dancing	Dance hall	264	249	89	160	49	35
Walking 3 mph; light machine work	Factory	293	293	110	183		
Bowling[d]	Bowling alley	440	425	170	255		
Heavy work	Factory	440	425	170	255	54	19
Heavy machine work; lifting	Factory	469	469	186	283		
Athletics	Gymnasium	587	528	208	320		

Notes:

1. Tabulated values are based on 23.9°C room dry-bulb temperature. For 26.7°C room dry bulb, total heat remains the same, but sensible heat values should be decreased by approximately 20%, and latent heat values increased accordingly.
2. Also see Table 4, Chapter 9, for additional rates of metabolic heat generation.
3. All values are rounded to nearest watt.

[a]Adjusted heat gain is based on normal percentage of men, women, and children for the application listed, and assumes that gain from an adult female is 85% of that for an adult male, and gain from a child is 75% of that for an adult male.

[b]Values approximated from data in Table 6, Chapter 9, where *V* is air velocity with limits shown in that table.

[c]Adjusted heat gain includes 18 W for food per individual (9 W sensible and 9 W latent).

[d]Figure one person per alley actually bowling, and all others as sitting (117 W) or standing or walking slowly (161 W).

Official Open

Cooling Load due to Lighting

- Ultimately released as heat.
- Storage effect. If the air conditioning is only operated when the lights are on, the cooling load factor is 1.00.
- Conventional ballast - add 20% (BF = 1.2) , electronic ballast - add 8 - 10% (BF = 1.08 – 1.1).

Official Open

Cooling Load due to Lighting

- Q_{c7} = A x LPD x CLF x DF or
  = W x BF x CLF x DF Watts

Where: A = Floor area
W = Total lamp wattage
LPD = Lighting power density (W/m^2)
BF = Ballast factor
DF = Diversity Factor

SS 530 - Maximum Lighting Power Density

Type of usage		Maximum lighting power density (Watts/m^2)
Offices, meeting rooms, copy/print rooms, classrooms, lecture theatres, computer rooms, reading rooms		12
Laboratories		16
Entrance halls, atriums, concourses, lobbies, auditoriums, concert halls		10
Retail general lighting and (accent, display, decorative)	Jewelry & crystal	15 + (20)
	Furniture, clothing & accessories, cosmetics, artwork	15 + (10)
	Others e.g. supermarket, vehicles, sporting goods, hardware, stationery, pharmacy and undetermined usage during early design	15 + (5)
Food courts, canteens, hawker centres (including dining area and food preparation)		10

Official Open

SS 530 - Maximum Lighting Power Density

Restaurants, lounges, bars (including dining area and food preparation, accent, display, decorative)	12
Circulation areas, corridors	7
Stairs, escalators, travelators	6
Lift lobbies	7
Car parks	3
Loading docks	5
Warehouses	7
Storage areas	10
Hotel guest rooms (including accent, decorative lighting, including spaces within the guest room such as toilets)	12
Toilets, changing rooms, laundries, washing areas	10
Gymnasiums and physical exercise areas	11
Mechanical and electrical rooms	10
Manufacturing (low to extra-high bays)	13
Manufacturing - electronic manufacturing and fine detail/Assembly industries	14

Official Open

ASHRAE - Lighting Power Density

Table 2 Lighting Power Densities Using Space-by-Space Method

Common Space Types*	LPD, W/m²	Common Space Types[a]	LPD, W/m²	Building-Specific Space Types*	LPD, W/m²
Atrium		**Lobby**		Playing area	9.2
<6.1 m in height	4.6	Facility for the visually impaired (and not used primarily by staff)[c]	18.2	**Health Care Facility**	
≥6.1 m and ≤12.2 m in height	5.2			In exam/treatment room	15.1
>12.2m in height	6.5	Elevator	7.0	In imaging room	10.1
Audience Seating Area		Hotel	5.4	Medical supply room	6.7
Auditorium	6.5	Motion picture theater	2.5	Nursery	9.9
Gymnasium	2.5	Performing arts theater	13.5	Nurses' station	12.6
Motion picture theater	2.9	All other lobbies	9.0	Operating room	24.3
Penitentiary	7.2	**Locker Room**	5.6	Patient room	7.3
Performing arts theater	12.5	**Lounge/Breakroom**		Physical therapy room	9.8
Religious building	7.8	Health care facility	4.5	Recovery room	13.5
Sports arena	3.5	All other lounges/breakrooms	6.3	**Library**	
All other audience seating areas	2.5	**Office**		Reading area	10.3
Banking Activity Area	6.5	Enclosed and ≤23.2 m²	8.0	Stacks	12.7
Breakroom (*See* Lounge/Breakroom)		Enclosed and >23.2 m²	7.1	**Manufacturing Facility**	
Classroom/Lecture Hall/Training Room		Open plan	6.6	Detailed manufacturing area	8.6
Penitentiary	9.5	**Parking Area, Interior**	1.6	Equipment room	8.2
All other classrooms/lecture halls/ training rooms	7.6	**Pharmacy Area**	17.9	Extra-high-bay area (15.2 m floor-to-ceiling height)	15.3
Conference/Meeting/Multipurpose Room	10.5	**Restroom**		High-bay area (7.6 to 15.2 m floor-to-ceiling height)	13.4
		Facility for the visually impaired (and not used primarily by staff)[c]	13.5		
Confinement Cells	7.5	All other restrooms	6.8	Low-bay area (<7.6 m floor-to-ceiling height)	9.3
Copy/Print Room	3.3	**Sales Area[d]**	11.3		
Corridor[b]		**Seating Area, General**	2.5	**Museum**	
Facility for visually impaired (and not used primarily by staff)[c]	7.7	**Stairway**		General exhibition area	3.3
		Space containing stairway determines LPD and control requirements for stairway.		Restoration room	11.9
Hospital	7.7			**Performing Arts Theater, Dressing Room**	4.4
All other corridors	4.4	**Stairwell**	5.3		

Official Open

ASHRAE - Lighting Power Density

Building-Specific Space Types*	LPD, W/m²
Courtroom	12.9
Computer Room	10.1
Dining Area	
Penitentiary	4.5
Facility for visually impaired (and not used primarily by staff)[c]	13.7
Bar/lounge or leisure dining	9.3
Cafeteria or fast food dining	4.3
Family dining	6.5
All other dining areas	4.7
Electrical/Mechanical Room[f]	4.6
Emergency Vehicle Garage	5.6
Food Preparation Area	11.7
Guest Room	4.4
Laboratory	
In or as classroom	11.9
All other laboratories	14.3
Laundry/Washing Area	5.7
Loading Dock, Interior	9.5
Storage Room	
<4.6 m²	5.5
>4.6 m²	4.1
Vehicular Maintenance Area	6.5
Workshop	13.5
Facility for Visually Impaired[c]	
Chapel (used primarily by residents)	7.5
Recreation room/common living room (and not used primarily by staff)	19.0
Automotive (*See* **Vehicular Maintenance Area**)	
Convention Center: Exhibit Space	6.6
Dormitory/Living Quarters	5.4
Fire Station: Sleeping Quarters	2.5
Gymnasium/Fitness Center	
Exercise area	9.6
Post Office, Sorting Area	8.1
Religious Buildings	
Fellowship hall	5.8
Worship/pulpit/choir area	9.2
Retail Facilities	
Dressing/fitting room	5.4
Mall concourse	8.8
Sports Arena, Playing Area[g]	
Class I facility	31.6
Class II facility	21.6
Class III facility	13.9
Class IV facility	9.3
Transportation Facility	
Baggage/carousel area	4.2
Airport concourse	2.7
Ticket counter	5.5
Warehouse—Storage Area	
Medium to bulky, palletized items	3.6
Smaller, hand-carried items[e]	7.4

Source: ASHRAE *Standard* 90.1-2019, Table 9.6.1.

[a]In cases where both a common space type and a building-specific type are listed, the building-specific space type applies.

[b]In corridors, the extra lighting power density allowance is granted when corridor width is <2.4 m and is not based on room/corridor ratio (RCR).

[c]A "Facility for the Visually Impaired" is a facility that can be documented as being designed to comply with light levels in ANSI/IES RP-28 and is licensed or will be licensed by local/state authorities for either senior long-term care, adult daycare, senior support, and/or people with special visual needs.

[d]For accent lighting, see section 9.6.2(b) of ASHRAE *Standard* 90.1-2019.

[e]Sometimes called a "Picking Area".

[f]An additional 5.6 W/m² shall be allowed, provided that the additional lighting is controlled separately from the base allowance of 4.6 W/m². The additional 5.6 W/m² allowance shall not be used for any other purpose.

[g]Class of play as defined by IES RP-6.

Official Open

Cooling Load due to Equipment

- Q_{c8} = Total power of running equipment x DF or
  = A x LF

Where: A = floor area (m^2)
LF = Load Factor (W/m^2)

Obtain specific data wherever possible.

Official Open

Recommended heat gain from computers, monitors, laser printers & copiers

Description	Continuous, W	Energy Saver Mode/Idle, W
Computers		
Average value	55	20
Conservative value	65	25
Highly conservative value	75	30
Monitors		
Small monitor (330 to 380mm)	55	0
Medium monitor (400 to 460 mm)	70	0
Large monitor (48 to 510 mm)	80	0
Laser Printers		
Small desktop	130	10 (Idle)
Desktop	215	35 (Idle)
Small office	320	70 (Idle)
Large office	550	125 (Idle)
Copier		
Desktop copier	400	20(Idle)
Office copier	1,100	300(Idle)

Official Open

Recommended heat gain from computers, monitors, laser printers & copiers

Description	Continuous, W	Energy Saver Mode/Idle, W
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Average value	55	20
Conservative value	65	25
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Monitors		
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Large monitor (48 to 510 mm)	80	0
Laser Printers		
Small desktop	130	10 (Idle)
Desktop	215	35 (Idle)
Small office	320	70 (Idle)
Large office	550	125 (Idle)
Copier		
Desktop copier	400	20(Idle)
Office copier	1,100	300(Idle)

Official Open

Recommended Equipment Load Factors for Various Types of Offices

Type of Use	Load Factor (W/m^2)	Description
100% laptop		
Light	2.69	15.5 m^2/workstation, all laptop use, 1 printer per 10, speakers, misc.
Medium	3.55	11.6 m^2/workstation, all laptop use, 1 printer per 10, speakers, misc.
50% laptop		
Light	4.31	15.5 m^2/workstation, 50% laptop/50% desktop, 1 printer per 10, speakers, misc.
Medium	5.38	11.6 m^2/workstation, 50% laptop/50% desktop, 1 printer per 10, speakers, misc.

Official Open

Recommended Equipment Load Factors for Various Types of Offices

100% desktop		
Light	6.46	15.5 m^2/workstation, all desktop use, 1 printer per 10, speakers, misc.
Medium	8.61	11.6 m^2/workstation, all desktop use, 1 printer per 10, speakers, misc.
100% desktop, 2 monitors	10.76	11.6 m^2/workstation, all desktop use, 2 monitors, 1 printer per 10, speakers, misc.
100% desktop, Heavy	16.15	7.9 m^2/workstation, all desktop use, 2 monitors, 1 printer per 8, speakers, misc.
100% desktop, Full on	21.5	7.9 m^2/workstation, all desktop use, 2 monitors, 1 printer per 8, speakers, misc., no diversity

Official Open

Recommended Equipment Load Factors for Various Types of Offices

Table 11 Recommended Load Factors for Various Types of Offices

Type of Use	Load Factor,* W/m^2	Description
100% laptop, docking station		
light	3.67	15.5 m^2/workstation, all laptop docking station use, 1 printer per 10
medium	4.91	11.6 m^2/workstation, all laptop docking station use, 1 printer per 10
50% laptop, docking station		
light	4.75	15.5 m^2/workstation, 50% laptop docking station/50% desktop, 1 printer per 10
medium	6.35	11.6 m^2/workstation, 50% laptop docking station/50% desktop, 1 printer per 10
100% desktop		
light	5.83	15.5 m^2/workstation, all desktop use, 1 printer per 10
medium	7.79	11.6 m^2/workstation, all desktop use, 1 printer per 10
100% laptop, docking station		
2 screens	7.44	11.6 m^2/workstation, all laptop docking station use, 2 screens, 1 printer per 10
100% desktop		
2 screens	9.06	11.6 m^2/workstation, all laptop use, 2 screens, 1 printer per 10
3 screens	10.33	11.6 m^2/workstation, all desktop use, 3 screens, 1 printer per 10
100% desktop		
heavy, 2 screens	11.00	7.9 m^2/workstation, all desktop use, 2 screens, 1 printer per 8
heavy, 3 screens	12.49	7.9 m^2/workstation, all desktop use, 3 screens, 1 printer per 8
100% laptop, docking station		
full on, 2 screens	12.23	7.9 m^2/workstation, all laptop docking use, 2 screens, 1 printer per 8, no diversity
100% desktop		
full on, 2 screens	14.35	7.9 m^2/workstation, all desktop use, 2 screens, 1 printer per 8, no diversity
full on, 3 screens	16.48	7.9 m^2/workstation, all desktop use, 3 screens, 1 printer per 8, no diversity

Source: Bach and Sarfraz (2018)
*Medium office type monochrome printer is used for load factor calculator with 15- min peak power consumption of 142 W.

Official Open

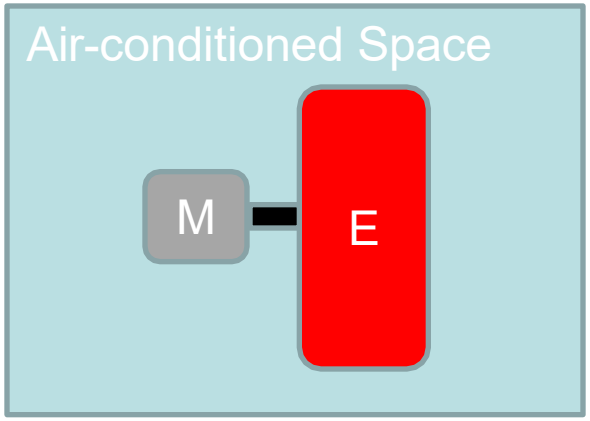
Cooling Load due to Electric Motors

Motor, driven equipment and useful output all situated within space.

$$\text{Cooling Load} = P_i$$

Where: P_i = power input to motor

η_m = motor efficiency

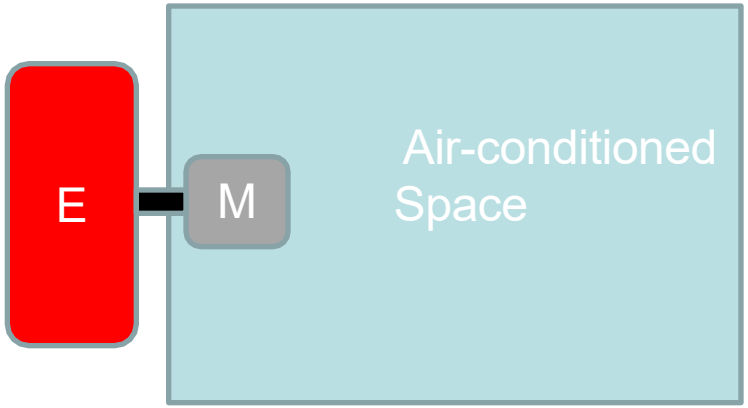


Official Open

Cooling Load due to Electric Motors

Motor situated within space connected with driven equipment outside space.

$$\text{Cooling Load} = P_i.(1 - \eta_m)$$



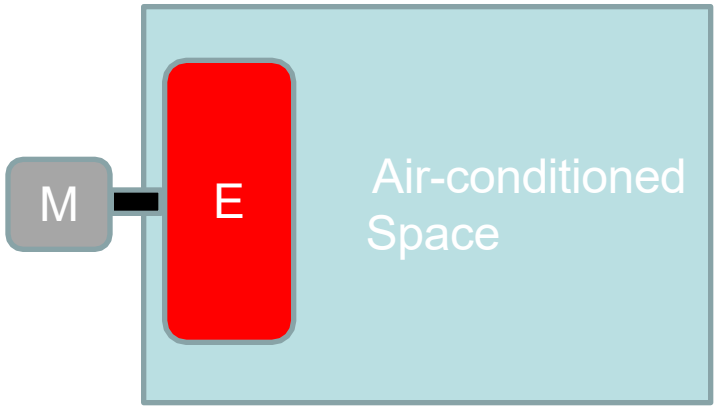
Official Open

Cooling Load due to Electric Motors

Driven equipment within space connected with motor outside space.

$$\text{Cooling Load} = P_i.\eta_m$$

Loss due to drive system (e.g. coupling, gearbox or pulley & belt) assumed to be negligible.



Official Open

Cooling Load due to Infiltration

- Infiltration dependent on wind pressure, stack effect and opening of doors.
- Natural ventilation is airflow resulting from designed provision of specified apertures.
- Both not constant.
- From experiments, median infiltration rate is 0.5 air change per h

Official Open

Space

Cooling Load due to Infiltration

- Sensible Cooling Load = $m_a.c_{pa}.(t_o - t_i)$ W

$$Q_{c9} = \frac{nV}{3600} \times \frac{1}{v_a} \times c_{pa}(t_o - t_i) \quad W$$

- Latent Cooling Load = $m_a.h_{fg}.(g_o - g_i)$ W

$$Q_{c10}' = \frac{nV}{3600} \times \frac{1}{v_a} \times h_{fg}\,(g_o - g_i) \quad W$$

C_{pa} = specific heat of air at constant pressure, = 1.012 kJ/kgK

V = volume of air-conditioned space.

ν_a = specific volume of air at condition that volume airflow rate is expressed.

h_{fg} = latent heat of vaporisation of water at about 20°C, = 2454 kJ/kg

Official Open

Space (or Room) Cooling Load

Sum of various cooling loads

The sensible space cooling load

$= Q_{c1} + Q_{c2} + Q_{c3} + Q_{c4} + Q_{c5} + Q_{c7} + Q_{c8} + Q_{c9}$

The latent space cooling load

$= Q_{c6}' + Q_{c10}'$

Official Open

Space (or Room) Cooling Load

Total space cooling load

= sensible space cooling load +
latent space cooling load

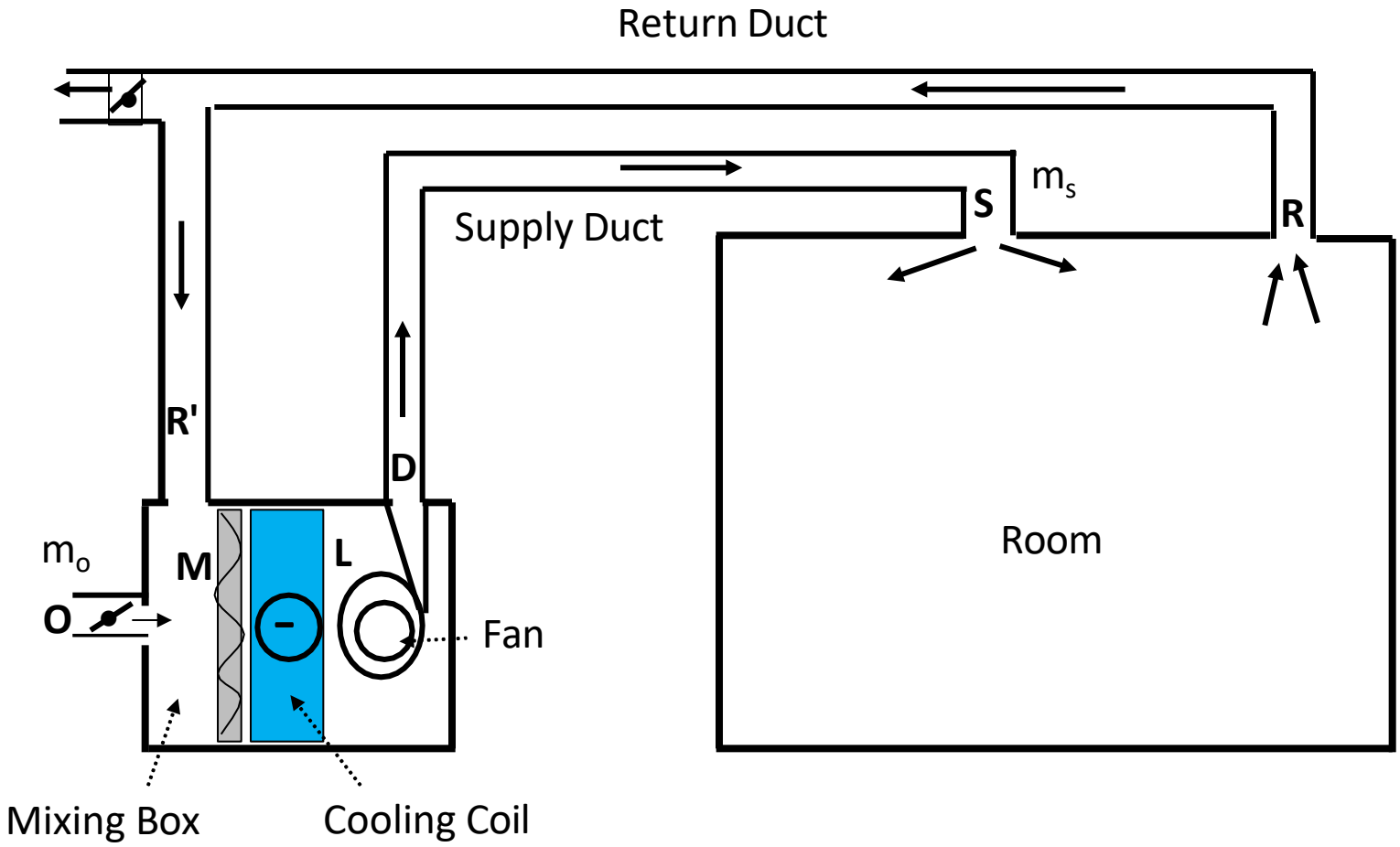
$= (Q_{c1} + Q_{c2} + Q_{c3} + Q_{c4} + Q_{c5} + Q_{c7} + Q_{c8} + Q_{c9}) + (Q_{c6}' + Q_{c10}')$

Official Open

External Cooling Loads

- Cooling Load due to Supply and Return Air Ducts
- Cooling Load due to Air Handling Fans
- Cooling Load due to Ventilation

Space Heat Gain & Cooling Load & Cooling Coil Load



Official Open

Cooling Load due to Supply and Return Air Ducts

- Heat gain due to supply air ducts in unconditioned space reduced cooling capacity of the plant and has to be compensated.
- In practice, cooling load due to supply air duct is about 1 to 4 % of space sensible cooling load.
- Cooling load due to return air duct normally neglected.
- Duct heat gain, Q_{duct} = y% x Space SCL Watts

Official Open

Cooling Load due to Supply and Return Air Ducts

- Alternatively, duct heat gain can be estimated from :

$$Q_{duct} = U_d \times A \times TD \quad \text{Watts}$$

Where: Q_{duct} = heat gain through duct walls (W)

U_d = thermal transmittance of duct (W/m^2K)

A = surface area of duct walls (m^2)

Official Open

Cooling Load due to Air Handling Fans

Due to:

- Inefficiency of the fan
- Air static and velocity pressures
- Heat generated by motor and drive
- Can estimate as % of space sensible cooling load. Varies from 1% up to 10% depending on fan pressure and temperature difference of the room and supply air.

Official Open

Cooling Load due to Air Handling Fans

Fan Static Pressure (Pa)	% to add to Space Sensible Cooling Load
250	2.5
500	5.0
1000	10.0

Fan heat gain, Q_{fan} = z% x Space SCL Watts

Official Open

SS 553 : 2016 – Outdoor Air Supply

Type of building/ Occupancy	Minimum outdoor air supply			
	l/s per m^2 floor area	m^3/h per m^2 floor area	l/s per person	Air class[1]
Restaurants	3.4	12.2	5.1	2
[(i)] Dance halls	7.0	25.0	10.5	1
Gym	3.8	13.6	13.2	2
Offices	0.6	2.0	5.5	1
[(ii)] Shops, supermarkets and department stores	1.1	3.8	5.5	1 or 2
Theatres and cinemas seating area	2.0	7.3	3.0	1
Lobbies and corridors	0.3	1.1	3.3	1
Concourses	1.1	4.0	3.3	1
[(iii)] Hotel guest rooms	15.0 l/s per room	54.0 m^3/h per room	5.5	1
[(iv)] Classrooms				
Primary school children and above	2.8	10.0	4.2	1
Childcare centres	2.8	10.0	8.4	1

Official Open

SS 553 : 2016 – Outdoor Air Supply

NOTE:

i) Dance halls refer to night clubs. The outdoor air supply in discotheques requires 50% more than that in dance halls.

ii) Air classification in occupancy with generation of internal contaminants such as barbershops, beauty and nail salons and pet shops is Class 2 and should be exhausted so that the spaces are maintained at negative pressure relative to the surrounding spaces.

iii) The use of higher outdoor air supply in hotel guest rooms stipulated in Table 1 should take precedence.

iv) The occupancy load for primary school children and above is 1.5 m^2 per person referred to in the Code of Practice for Fire Precautions in Buildings. The occupancy load for childcare centres is based on 3 m^2 per person referred to in the "Guide to setting up a childcare centre", the Ministry of Community Development, Youth & Sports.

Official Open

Ventilation Load (Outside Air Load)

- Sensible Cooling Load = $m_a.c_{p,a}.(t_o - t_i)$ W

$$Qc = \frac{V'}{v_a} c_{p,a}(t_o - t_i) \quad W$$

- Latent Cooling Load = $m_a.h_{fg}.(g_o - g_i)$ W

$$Qc' = \frac{V'}{v_a} h_{fg} (g_o - g_i) \quad W$$

V' = volume flow rate of the outside air, m^3/s

v_a = the specific volume of room air, m^3/kg

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Cooling Coil Load

- Selected to cater to the space cooling load as well as the external loads.
- Sensible Cooling Coil Load

  = Sensible space cooling load +
  Sensible external cooling load

Official Open

Cooling Coil Load

- Latent Cooling Coil Load

  = Latent space cooling load +
  Latent external cooling load

- Total Cooling Coil Load

  = Total space cooling load +
  Total external cooling load

Official Open

Diversity Factors in Cooling Load Estimation

- Diversity factor may be used for lighting, people and equipment
- Only apply after careful study and analysis of the probable situations.
- Fairly common not to apply any diversity factor.

Official Open

Design and Cooling Check Figures

Applications	Room Sensible Cooling Load, W/m²			Total Room Cooling Load, W/m²			Total Cooling Coil Load, W/m²			Supply Air Quantity, l/s/m²		
	Low	Ave	High	Low	Ave	High	Low	Ave	High	Low	Ave	High
Offices												
Private	79	158	237	95	174	252	126	237	284	5.0	8.5	12.0
Perimeter	63	110	221	79	126	237	95	158	268	5.0	6.0	11.5
Interior	47	63	95	63	79	110	79	95	126	3.8	5.0	5.5
Conference Room	95	174	252	126	205	284	189	268	379	5.0	9.0	13.5
Restaurants	95	110	158	126	158	221	189	268	379	6.3	7.5	10.0
Department Stores												
Basement	63	95	142	79	110	158	110	142	189	5.0	7.0	8.8
Ground Floor	79	110	142	95	126	158	126	158	189	5.0	7.5	10.0
Upper Floor	47	79	110	63	95	126	95	126	158	4.0	5.0	6.0

Official Open

Design and Cooling Check Figures

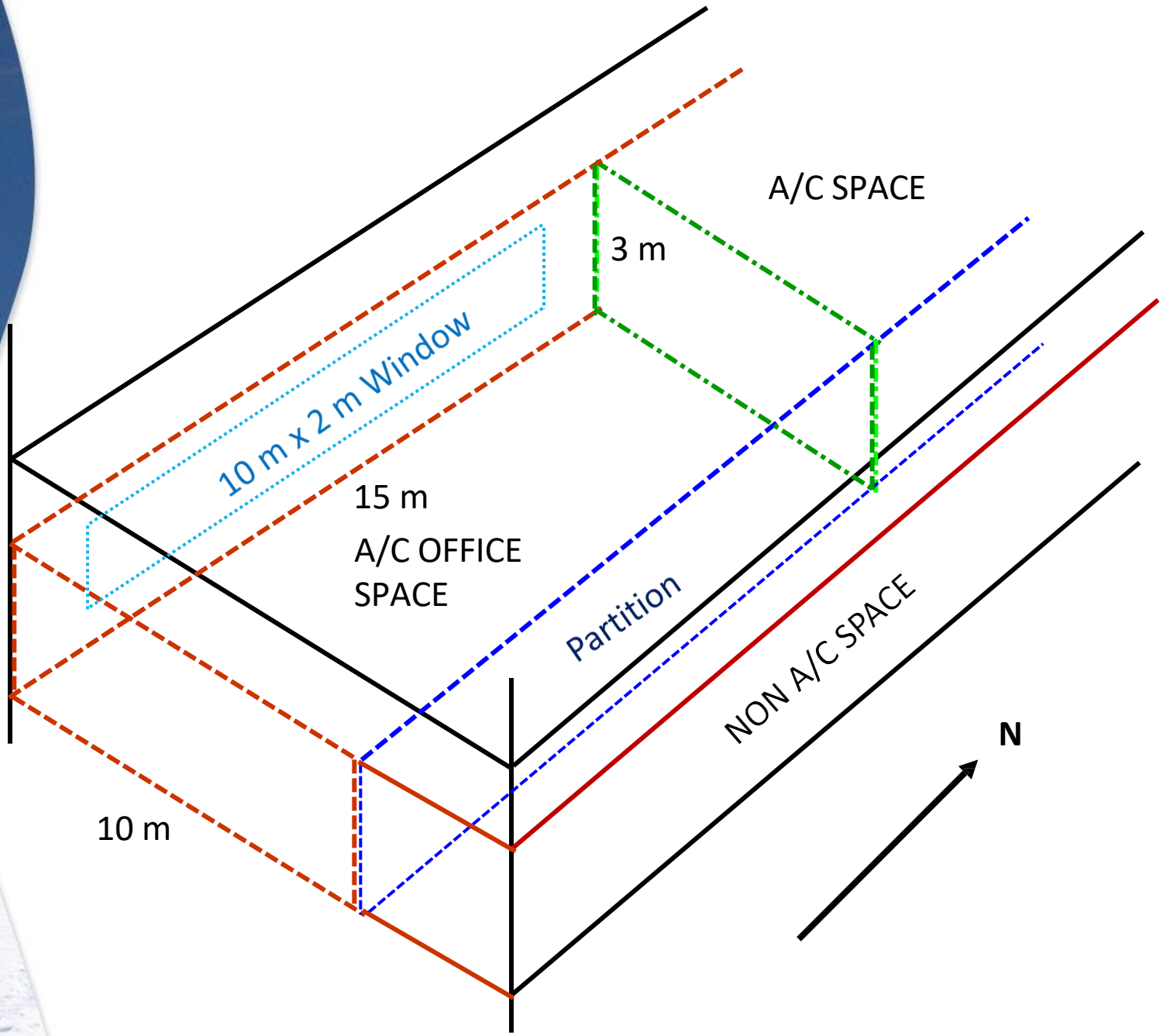
Applications	Room Sensible Cooling Load, W/m²			Total Room Cooling Load, W/m²			Total Cooling Coil Load, W/m²			Supply Air Quantity, l/s/m²		
	Low	Ave	High	Low	Ave	High	Low	Ave	High	Low	Ave	High
Educational Facilities												
Classroom	79	126	174	110	158	205	142	189	252	5.0	7.0	9.0
Laboratory	95	126	174	110	142	205	142	189	237	5.0	7.0	9.0
Hospital												
Patient Area	47	110	158	63	126	174	189	379	521	3.8	6.0	8.5
Public Area	32	47	110	47	63	126	95	142	316	3.8	6.0	8.5
Clinic	63	126	189	79	142	205	126	189	252	5.0	7.0	10.0
Theatre	79	110	158	142	174	221	189	252	379	6.3	7.5	12.5
Apartment	47	79	142	63	95	158	95	126	189	3.8	6.3	8.5

Official Open

Example – Cooling Load Estimation

Figure 1 shows the isometric view of an office space on the intermediate floor of a building in Singapore. Determine the space cooling load, space latent cooling load, space total cooling load and cooling coil load for the space based on the design information provided below.

Official Open



Official Open

Design Information

Month of Interest : March
Time of Interest : 16 h
Outside air condition : 33°C dbt, 27°C wbt, 8°C range
Inside air condition : 23°C db, 55% saturation
Building Weight : Medium

Window

Type of glazing : Uncoated, single glazing, 6.4 mm bronze
U-value of glazing : 5.6 W/m^2 K

Walls

Type of the external wall : Group D
U-value of the external wall : 2.5 W/ m^2 K

Official Open

Internal Walls/Partitions & Floor

There is no heat gain through the floor and ceiling.

U-value of the internal walls : 2.0 W/ m^2K

Temperature in unconditioned space : 30°C

Occupancy

Density of occupancy : 8 m^2/person

Activity : Office

Diversity Factor : 1.0

Lighting

Lighting load : Based on SS 530

Diversity Factor : 0.9

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Equipment

Equipment load : 50% laptop, medium

Infiltration

Rate of infiltration (air change/hr): 0.3

Ventilation : Based on SS 553

Supply duct heat gain : 2% of Space sensible cooling load

Estimated fan static pressure : 300 Pa

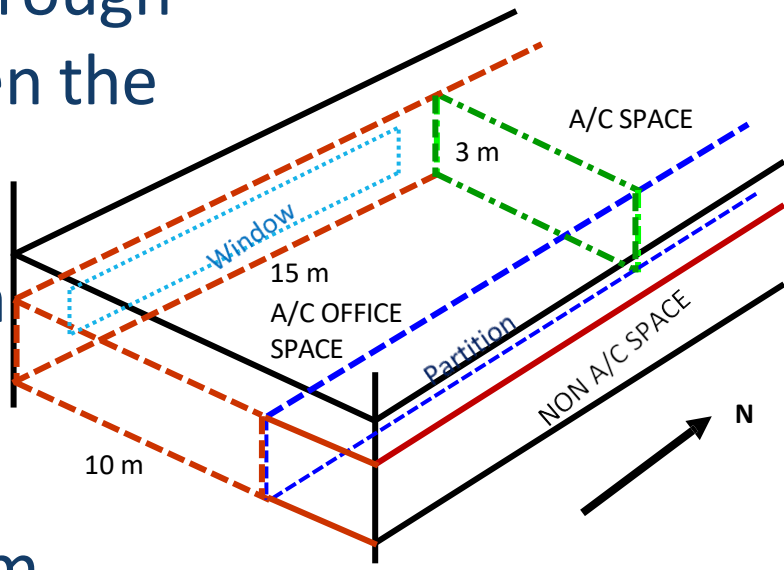
Official Open

Example

Find sensible cooling load thru' through window due to solar radiation given the following:

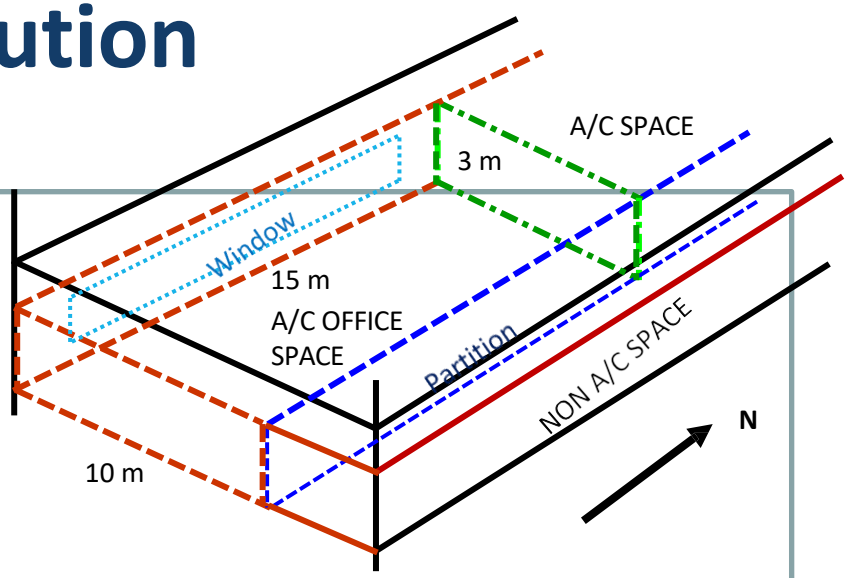
Window dimensions : 10 x 2m
Month of Interest : March
Time of Interest : 16 h
Building Weight : Medium
Type of glazing : Uncoated, single glazing, 6.4 mm bronze

Window orientation : West



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Solution



$Q_{c1} = A_g \times SHGF \times SC \times CLF$

$A_g = 10 \times 2 = 20\ m^2$

From Table 4.1, Mar, West, SHGF = 755 W/m^2

From Table 4.3, West, Medium, 16 h, CLF = 0.87

From Table 4.2, Uncoated, single glazing, 6.4 mm bronze, SC = 0.55

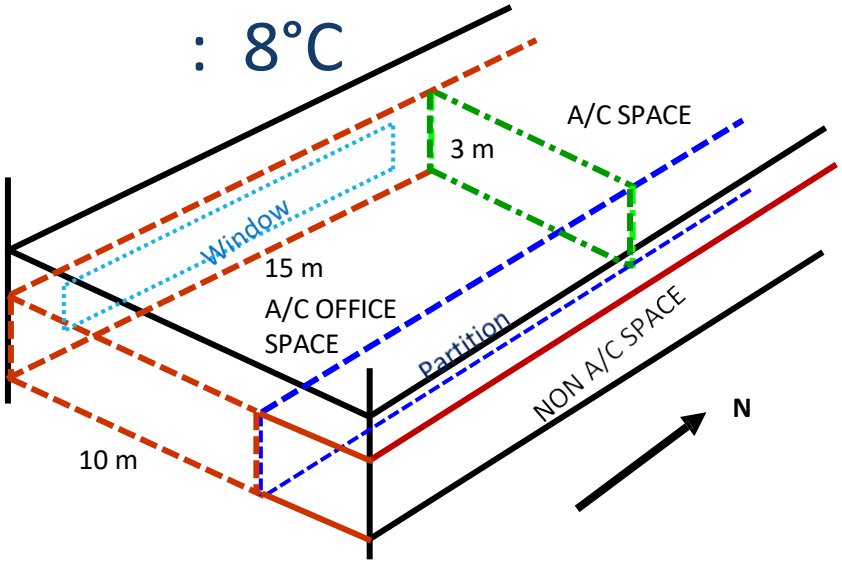
$Q_{c1} = 20 \times 755 \times 0.55 \times 0.87 = 7225\ W$

Official Open

Example

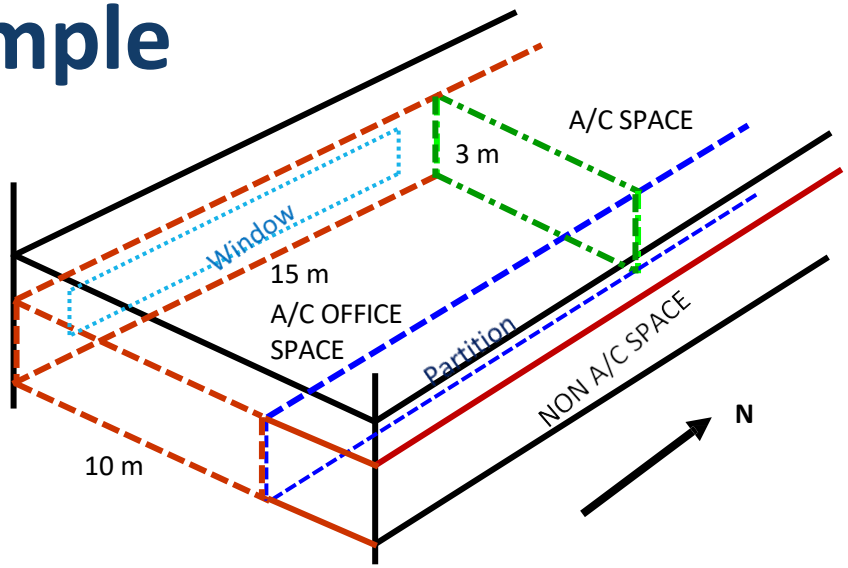
- Find heat gain through window due to transmission given as follows:

Window dimensions	: 10 x 2m
U-value	: 5.6 W/ m^2 K
A/C room air temp	: 23°C
Outside air design temp	: 33°C
Daily range	: 8°C



Official Open

Example



Solution

$Q_{c2} = A_g \times U_g \times CLTD$

Corrected CLTD $= CLTD + (25.5 - t_r) + (t_m - 29.4)$

From Table 4.4, 16 h, CLTD $= 8°C$

Corrected CLTD $= 8 + (25.5 - 23) + (33 - (8 \div 2) - 29.4)$
$= 10.1°C$

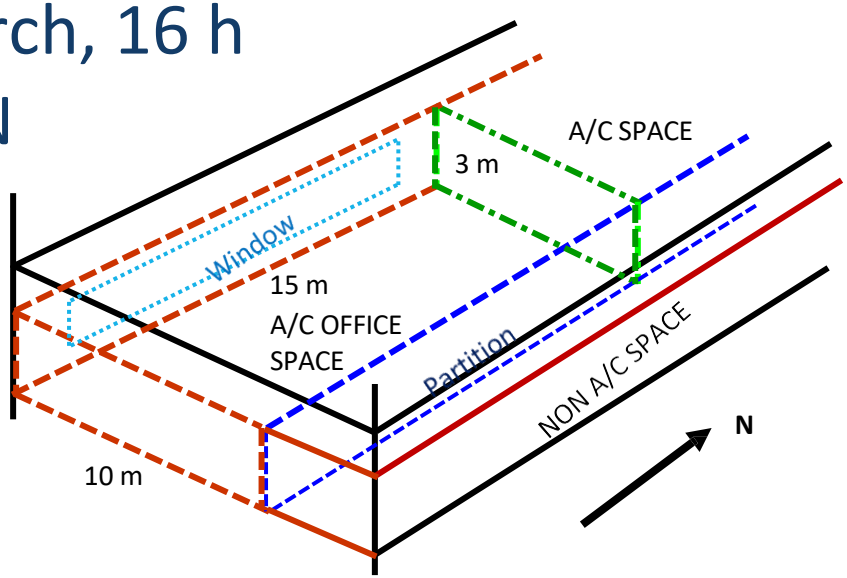
$Q_{c2} = 20 \times 5.6 \times 10.1 = 1131$ W

Official Open

Example

Find heat gain thru' an opaque wall given the following:

Wall dimensions : 15 x 3m (window 10 x 2m)
Type of Wall : Group D
U-value : 2.5 W/ m^2 K
Wall orientation : West
Month and time : March, 16 h
Latitude : 0 °N



Space

Solution

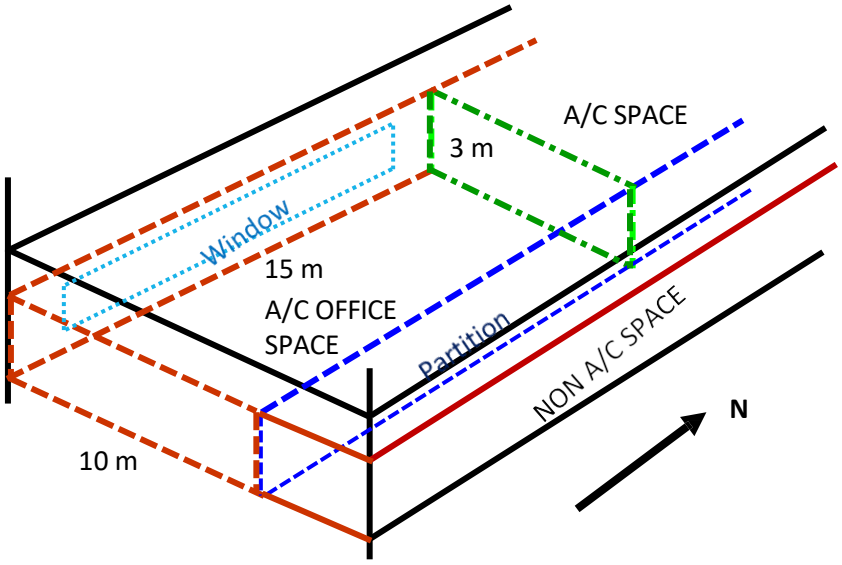
$Q_{c3} = A_w \times U_w \times CLTD$ Watts

West facing wall

$A_w = (15 \times 3 - 10 \times 2) = 25\ m^2$, $U_w = 2.5\ W/m^2K$

From Table 4.5 (a), West, 16 h, CLTD = 8.3°C

$Q_{c3W} = 25 \times 2.5 \times 8.3 = 519\ W$

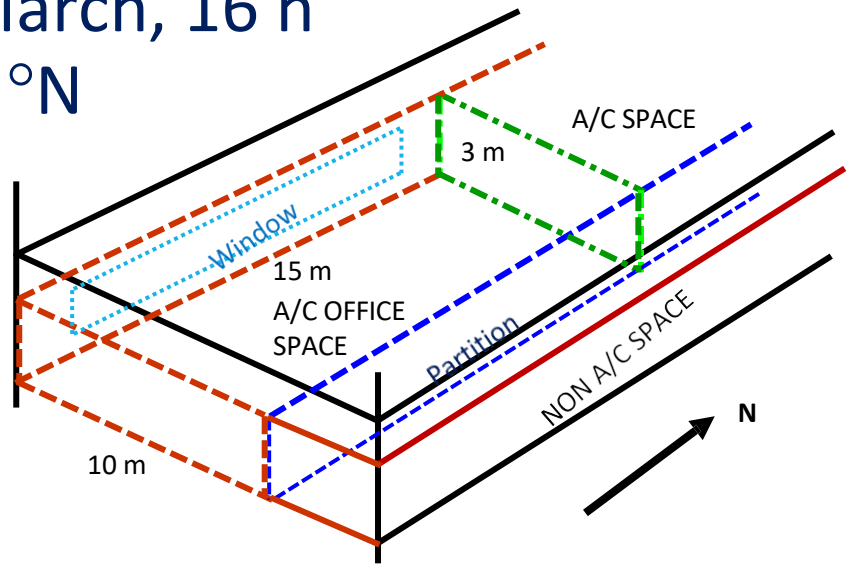


Official Open

Example

Find heat gain thru' an opaque wall given the following:

Wall dimensions : 10 x 3m
Type of Wall : Group D
U-value : 2.5 W/ m^2 K
Wall orientation : South
Month and time : March, 16 h
Latitude : 0 °N



Space

Solution

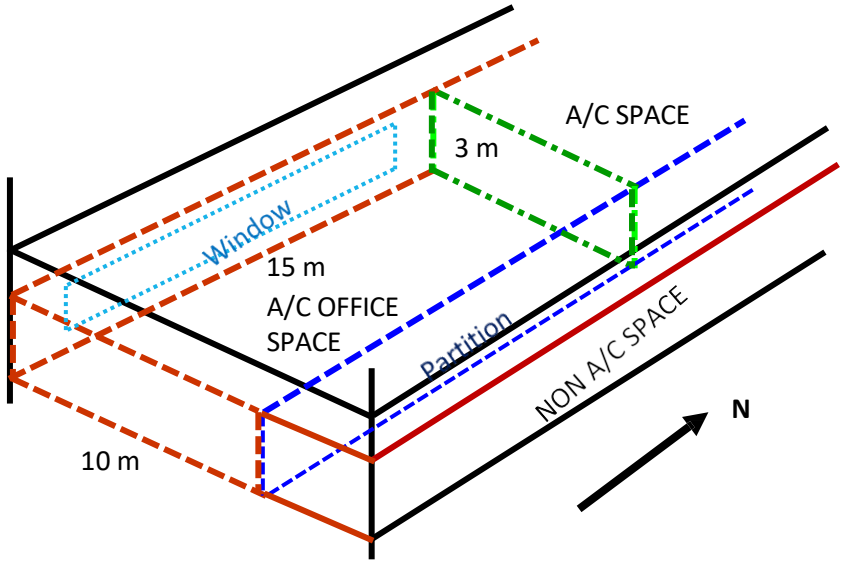
$Q_{c3} = A_w \times U_w \times CLTD$ Watts

South facing wall

$A_w = (10 \times 3) = 30\ m^2$, $U_w = 2.5\ W/m^2K$

From Table 4.5 (a), South, 16 h, CLTD = 8.9°C

$Q_{c3S} = 30 \times 2.5 \times 8.9 = 668\ W$

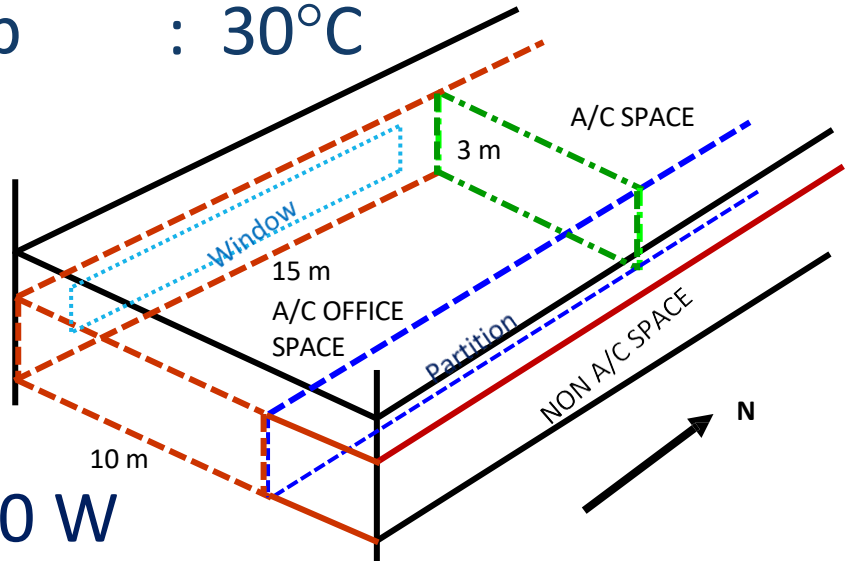


Official Open

Example

Find heat gain thru' an internal partition given the following:

Partition dimensions	: 15 x 3m
U-value	: 2.0 W/ m^2K
A/C room air temp	: 23°C
Non-A/C space temp	: 30°C



Solution

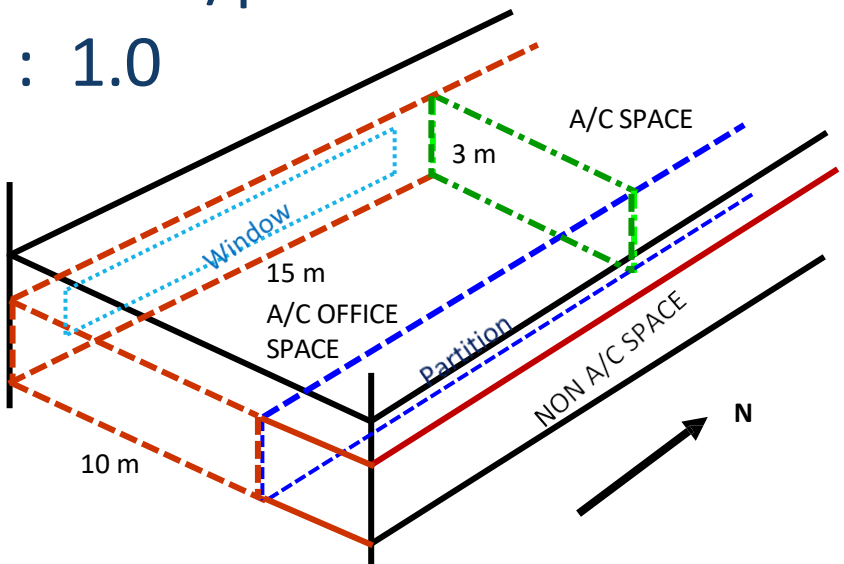
$Q_{c4} = A_w.U_w.(t_{nr} - t_i)$

$Q_{c4} = 15x3 \times 2.0 \times (30 - 23) = 630$ W

Example

Find heat gain due to occupancy given the following:

Space dimensions : 15 x 10 m
Application : Office
Space dry bulb temp. : 23°C
Occupancy density : 8 m^2/person
Diversity Factor : 1.0



Official Open

Space

Solution

Q_{c5} = No. of occ. x sensible gain/person x CLF x DF

Q_{c6}' = No. of occ. x latent gain/person x CLF x DF

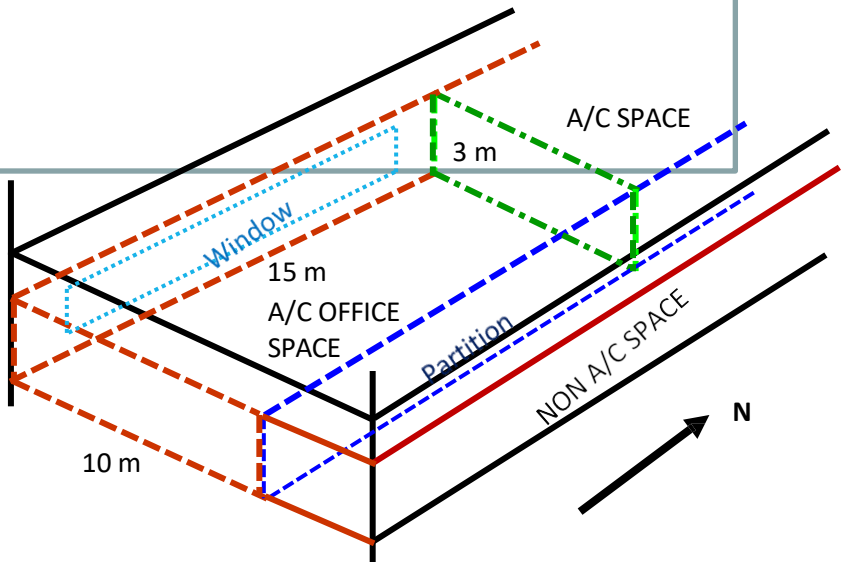
No. of occupants = Floor area/ Occupancy density

= (15 x 10)/8

= 19

From Table 4.6, Office, 23°C, SHG/person = 85 W

LHG/person = 55 W



Q_{c5} = 19 x 85 x 1.0 x 1.0 = 1615 W

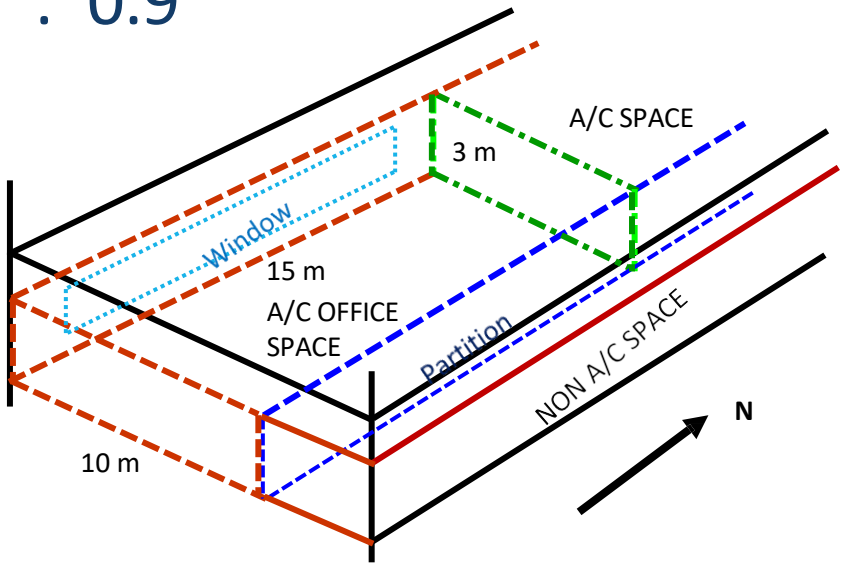
Q_{c6}' = 19 x 55 x 1.0 x 1.0 = 1045 W

Official Open

Example

Find heat gain thru' due to lighting given the following:

Space dimensions : 15 x 10 m
Application : Office
Type of lighting : Fluorescent
Diversity Factor : 0.9



Official Open

Solution

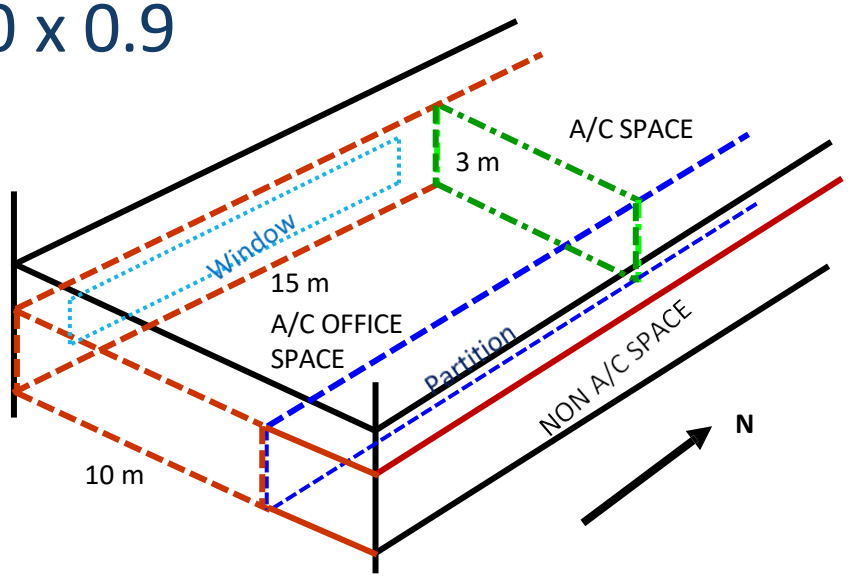
Q_{c7} = A x LPD x CLF x DF

A = 15 x 10 = 150 m^2

From Table 4.7, office,
LPD = 12 W/m^2

Q_{c7} = 150 x 12 x 1.0 x 0.9

= 1620 W



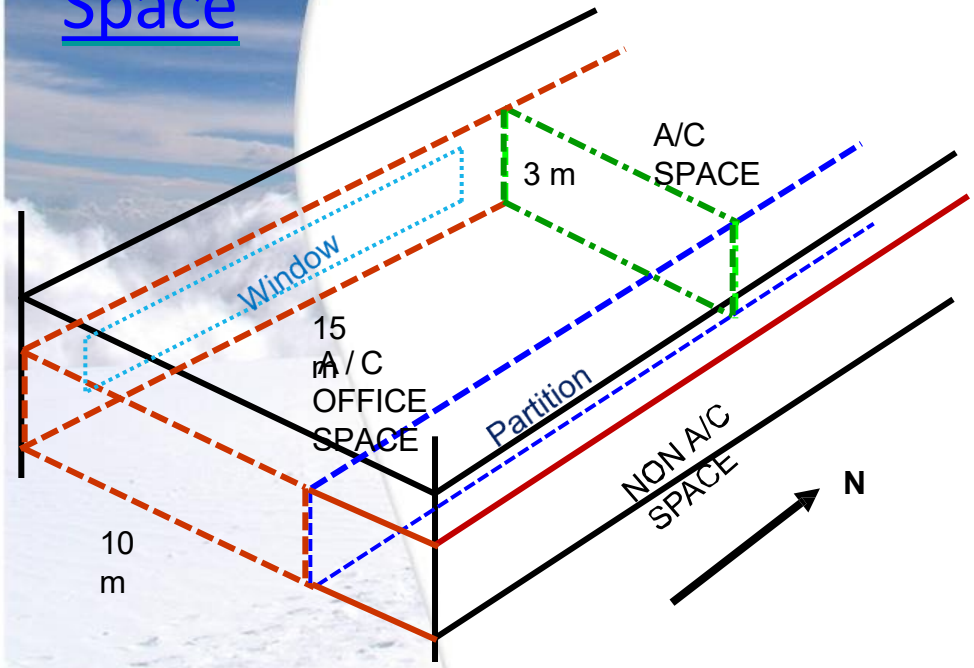
Official Open

Example

Find heat gain due to equipment given the following:

Space dimensions : 15 x 10 m
Application : Office
Load Density : 50% laptop, medium

Space



Solution

From Table 4.9, 50% laptop, medium, LF = 5.38 W/m^2

$Q_{c8} = A \times LF$

$= 150 \times 5.38$

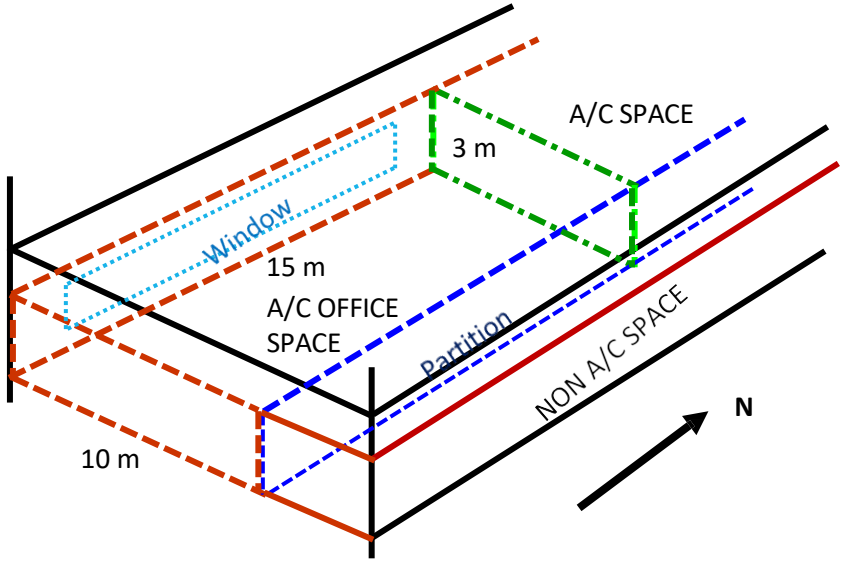
$= 807$ W

Official Open

Example

Determine the heat gain due to infiltration given the following:

Space Dimensions : 15 x 10 x 3 m
Infiltration Rate : 0.3 ACH
Outside air condition : 33°C db, 27°C wb
Inside Air Condition : 23°C db, 55% satn



Official Open

No. of air change

Solution

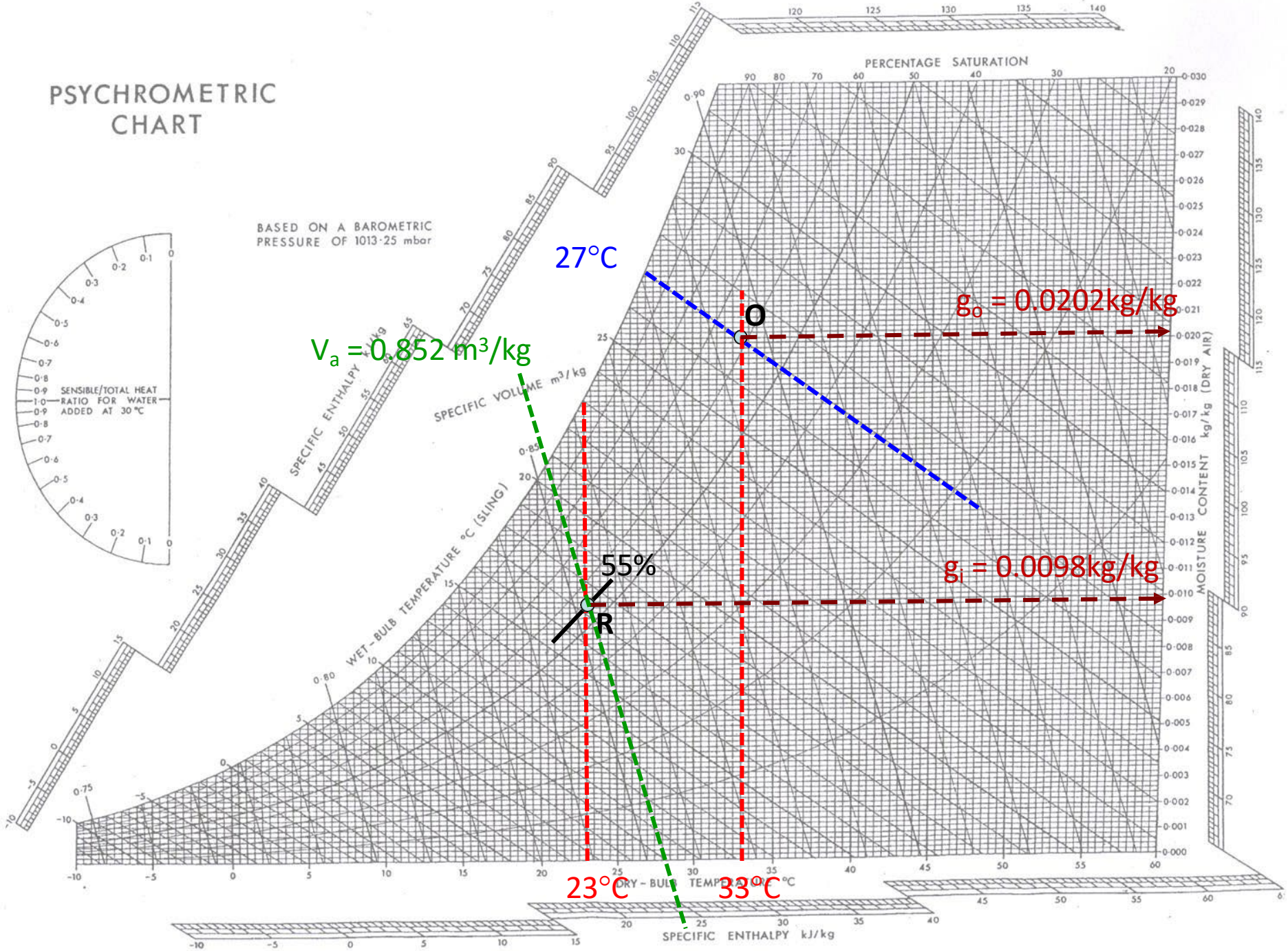
Room volume

$$Q_{c9} = \frac{nV}{3600} \times \frac{1}{v_a} \times c_{pa}(t_o - t_i) \quad W$$

$$Q_{c10}' = \frac{nV}{3600} \times \frac{1}{v_a} \times h_{fg}(g_o - g_i) \; W$$

Plot inside and outside conditions onto the psychrometric chart.

From plot, $\nu_a = 0.852 \; m^3/kg$, $g_i = 0.0098 \; kg/kg$ and $g_o = 0.0202 \; kg/kg$


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30 °C
PERCENTAGE SATURATION
SPECIFIC ENTHALPY kJ/kg
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE °C (SLING)
MOISTURE CONTENT kg/kg (DRY AIR)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
27°C
V_a = 0.852 m³/kg
O
g_o = 0.0202kg/kg
55%
R
g_i = 0.0098kg/kg
23°C
33°C

Official Open

Solution

$$Q_{c9} = \frac{0.3(15x10x3)}{3600} \times \frac{1}{0.852} \times (1.012x10^3)(33 - 23)$$

$$= 445 \text{ W}$$

$$Q_{c10}' = \frac{0.3(15x10x3)}{3600} \times \frac{1}{0.852} \times (2454x10^3)(0.0202 - 0.0098)$$

$$= 1123 \text{ W}$$

Official Open

Space (or Room) Cooling Load

The sensible space cooling load

= 7225 + 1131 + 519 + 668 + 630 + 1615 + 1620 + 807 + 445 = 14,660 W

The latent space cooling load

= 1045 + 1123 = 2168 W

Total space cooling load = 14,660 + 2,168

= 16,828 W

Space cooling load/m^2 = 16,828 ÷ (15x10)

= 112.19 W/m^2

Official Open

Supply Air Quantity

Assume that the supply air temperature is 13°C.

Supply air quantity

$$= \frac{\text{Sensible space cooling load}}{C_{p,a} \times (t_i - t_s)} \quad \text{kg/s}$$

$$= 14{,}660 \div (1.012 \times 10^3 \times (23 - 13))$$

$$= 1.449 \text{ kg/s}$$

Official Open

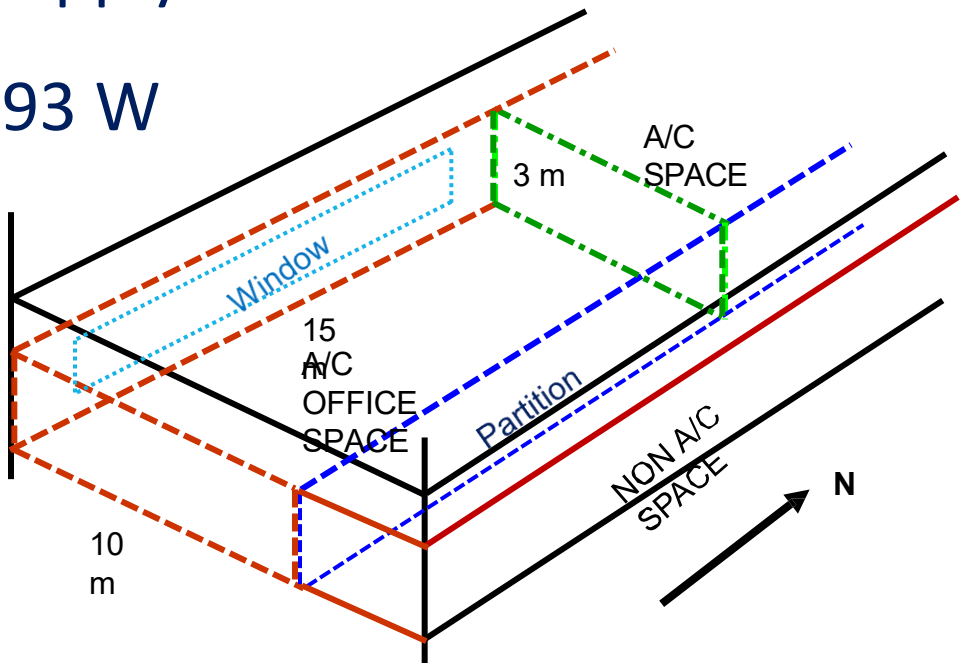
Example

Determine the duct heat gain given the following:

Sensible Cooling Load : 14,660 W
Supply duct heat gain : 2% of SCL

Heat gain due to supply duct

= 2% x 14,660 = 293 W



Official Open

Example

Determine the heat gain due to the AHU fan:

Estimated fan static pressure : 300 Pa
Space cooling load : 14,660 W

From Table, by interpolation at 300 Pa,
% of sensible cooling load = 3.0%

Heat gain due to fan

= 3.0% x 14,660 = 440 W

Fan Static Pressure (Pa)	% to add to Space Sensible Cooling Load
250	2.5
500	5.0
1000	10.0

Official Open

Example

Determine the heat gain due to ventilation air given the following based on SS 553:2016:

Space dimensions : 15 x 10 m
Application : Office
Outside air condition : 33°C db, 27°C wb
Inside Air Condition : 23°C db, 55% satn

Official Open

Solution

$$Q_c = \frac{V'}{v_a} \times c_{p,a} \times (t_o - t_i) \quad W$$

$$Q_c' = \frac{V'}{v_a} \times h_{fg} \times (g_o - g_i) \quad W$$

From Table 4.10, office, O.A supply = 0.6 l/s/m^2

$$\therefore V' = \frac{0.6}{1000} \times (15 \times 10) = 0.09 \text{ m}^3/\text{s}$$

$$Q_c = \frac{0.09}{0.852} \times 1.012 \times 10^3 (33 - 23) = 1069 \text{ W}$$

$$Q_c' = \frac{0.09}{0.852} \times 2454 \times 10^3 (0.0202 - 0.0098) = 2696 \text{ W}$$

Official Open

Cooling Coil Load

Total Cooling Coil Load

Total space cooling load		Duct heat gain		Fan heat gain		Ventilation load - sensible		Ventilation load - Latent
= 16,828	+	293	+	440	+	1069	+	2592

= 21,326 W

Cooling coil load/m^2 = 21,326 ÷ (15x10)

= 142.17 W/m^2

Quiz

Quiz 3
Cooling Load Estimation
19 questions amounting to 25 points
Three attempts. No backtracking, Auto-submission after 60 minutes
Attempt with highest final point will be taken as the final point.
Available from 0900 Friday, 30 May to 2359 Monday, 2 Jun 25

Tutorial AMVS 3

- Answer Questions 4 and 5.
- All answers shall be hand-written.
- Submission can be by hardcopy or through PoliteMall. For PoliteMall submission, the answers shall be scanned in only **ONE pdf file**
- Submission deadline : 2359 Thursday, 29 May 2025